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Turbulence modeling of a single-phase R134a supersonic ejector. Part 1: Numerical benchmark



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ABSTRACT

The present work reports a numerical analysis of a supersonic ejector in single-phase conditions using R134a as the working fluid. A numerical benchmark of some thermodynamic and two-equation turbulence models has been carried out to highlight the numerical model offering the best compromise between accuracy and calculation cost. The validation is achieved by comparing the predicted entrainment ratio with the experimental data of Garcia del Valle et al. (2014). The $k-\omega$ SST model together with the REFPROP 7.0 database equation appears to be the best combination to predict accurately the ejector performance and capture the shock wave structure. The influence of the outlet temperature, the discussion about the validity of some assumptions made by one-dimensional (1D) models and the exergy analysis within the ejector for the present operating conditions will later be discussed in Part 2 (Croquer et al., 2015).

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Modélisation de la turbulence d'un éjecteur supersonique au R134a monophasé. Partie 1: Référence numérique

Mots clés : Ejecteur supersonique ; Modélisation de la turbulence ; Frigorigène R134a ; Onde de choc

1. Introduction

The constant increase in energy consumption and decrease in natural resources necessitate a more efficient use of energy. Coupled with the political will to reduce the greenhouse gas emissions, many national initiatives develop to increase the

energy efficiency of industrial systems and replace refrigerants with a high environmental impact. In that context, a renewed interest has recently born to incorporate ejectors in refrigeration systems to replace either the expansion valve or the compressor. Besides their lack of moving parts and low maintenance requirements, supersonic ejectors can be driven by renewable or low quality energy sources, such as solar or

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单相R134a超音速喷射器的湍流建模.第一部分：数值基准



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摘要

本文报告了使用R134a作为工作流体的单相条件下超音速喷射器的数值分析。对某些热力学和双方程湍流模型进行了数值基准测试，以突出在精度和计算成本之间提供最佳折中的数值模型。通过将预测的引射比与Garcia del Valle等（2014）的实验数据进行比较来验证。结合REFPROP 7.0数据库方程的 $k-\omega$ SST模型似乎是准确预测喷射器性能并捕捉激波结构的最佳组合。出口温度的影响、一维（1D）模型的一些假设有效性的讨论以及当前操作条件下喷射器内的火用分析将在第二部分（Croquer等，2015）中讨论。

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单相R134a超音速喷射器的湍流建模。第一部分：数值基准

关键词：超音速喷射器；湍流建模；R134a制冷剂；激波

1. 引言

能源消耗的持续增长和自然资源的减少要求更高效地利用能源。结合减少温室气体排放的政治意愿，许多国家倡议正在发展以增加

提高工业系统的能效，并替换对环境有重大影响的制冷剂。在此背景下，最近重新燃起了将喷射器集成到制冷系统中以替代膨胀阀或压缩机的兴趣。除了没有运动部件和维护要求低之外，超音速喷射器可以利用可再生能源或低品位能源驱动，如太阳能或

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Nomenclature		δ	Film thickness (m)
c_p	Heat capacity ($J\ kg^{-1}\ K^{-1}$)	λ	Thermal conductivity ($W\ m^{-1}\ K^{-1}$)
D	Mass diffusivity ($m^2\ s^{-1}$)	μ	Dynamic viscosity (Pa s)
g	Gravitational acceleration ($m\ s^{-2}$)	ξ	Mass fraction (–)
h	Enthalpy ($J\ kg^{-1}$)	ρ	Density ($kg\ m^{-3}$)
$\bar{h}$	Partial massive enthalpy ($J\ kg^{-1}$)	Subscripts	
$\dot{m}$	Mass flux ($kg\ m^{-2}\ s^{-1}$)	o	Inlet condition
p	Pressure (Pa)	abs	Absorption
$\dot{q}$	Heat flux ($W\ m^{-2}$)	avg	Average
T	Temperature (K)	A	Absorbate e.g. water in liquid state
u	Longitudinal velocity ($m\ s^{-1}$)	des	Desorption
x	Longitudinal coordinate, streamwise (m)	eq	Equilibrium
y	Transversal coordinate (m)	if	Interface
Greek letters		s	Solution of absorbent e.g. LiBr – H ₂ O
$\dot{\Gamma}$	Mass flow per unit length ($kg\ m^{-1}\ s^{-1}$)	v	Vapour e.g. steam
Δh_{sor}	Heat of sorption ($J\ kg^{-1}$)	w	Wall

waste heat sources (Meyer et al., 2009). Recently, Bilir et al. (2015) studied the behavior of a vapor refrigeration cycle with an ejector as the expansion stage. Their exergy analysis of the whole system shows that the ejector expansion refrigeration cycle has a greater overall coefficient of performance by 7–12% over the conventional refrigeration cycle. Thus, their application is attractive from an economical, operational and environmental point of view to replace or support conventional compression–expansion devices (Sumeru et al., 2012). Ejectors may be also used to exploit low-pressure reservoirs, industrial vacuum generation, and for gas mixing inside breathing apparatus. The reader can refer to the reviews of Chunnanond and Aphornratana (2004), Elbel and Hrnjak (2008) and Elbel (2011) for a detailed picture of ejector systems in refrigeration or air-conditioning applications.

Numerous experimental or theoretical works have been published over the last decades considering a wide range of refrigerants, various ejector designs and for single and two-phase flows. The literature on supersonic ejectors is then too abundant to draw up an exhaustive state-of-art about the topic. The reader can refer to the review of Milazzo et al. (2014) for a recent picture on theoretical and experimental works on ejector refrigeration. In the following, the focus is on single-phase supersonic ejectors.

Given the small dimensions of the ejectors and the need of thermal insulation, experimental studies are frequently limited to global measurements: mass flowrates, temperature and pressure at the inlets and the outlet or more recently wall pressure measurements along the mixing chamber (Mazzelli and Milazzo, 2015). Attempts have been made by Bartosiewicz et al. (2005) to measure the pressure at the centerline of the ejector by invasive pressure probes. Only few works focused on the shock wave structure using local methods. Marinovski et al. (2009) investigated the droplet condensation phenomenon occurring in moist air ejector by laser tomography, giving also access to the shock train structure. Bouhanguel and Desevaux (2011) extended this work using different illumination sources, polarization directions and types of tracers to investigate specific flow features of a supersonic

ejector working with air. Zhu and Jiang (2014) studied the shock structure inside two kinds of ejectors by optical Schlieren measurements. They established the shock wave wavelength dependency on the primary inlet to outlet pressure ratio. They showed also that oblique shock waves appear when the ejector reaches the on-design condition, whereas normal shocks appear for off-design operation.

Numerical simulations appear then as a valuable tool to investigate the local flow features inside supersonic ejectors. Due to the relative geometrical complexity of ejectors and most of all to the very complex flow and energy transfer phenomena (shock waves, thin boundary layers, compressibility effects, phase change, etc.) involved in the problem, all authors focused on the physics of ejectors by 2D axisymmetric simulations using commercial softwares. Bartosiewicz et al. (2005) evaluated the performance of six two-equation turbulence models for single-phase supersonic ejectors. The k–ε RNG and SST models provided the best overall agreement compared to experimental data. It was partially confirmed by Zhu and Jiang (2014), which showed that the k–ε RNG model agrees better in terms of the shock prediction than the realizable and standard k–ε and SST models. They suggested a detailed modeling of the near-wall regions to achieve a good prediction of the flow structure inside the ejector. Scott et al. (2008, 2011) performed extensive calculations using the software Phoenix for a supersonic ejector working with R245fa. For various geometries and operating conditions, they obtained a very good agreement against experimental data in terms of the entrainment ratio with a maximum deviation of 10.8%. Cai and He (2013) studied the effect of different gas and turbulence models and geometry variations on the predictions of a steam supersonic ejector. The ideal gas model underpredicts the entrainment ratio by 20–40%. Recently, Mazzelli and Milazzo (2015) performed a combined experimental and numerical analysis of a supersonic ejector chiller with R245fa. Results show that the wall roughness has only a weak effect on the on-design entrainment ratio but can largely reduce the critical back pressure. The global ejector performance depends only slightly on the selected gas model. Pianthong et al. (2007) and

符号表		δ	薄膜厚度 (m)
c_p	热容 ($焦耳\ 千克^{-1}开尔文^{-1}$)	λ	导热系数 ($瓦\ 米^{-1}开尔文^{-1}$)
D	质量扩散率 ($米^2\ 秒^{-1}$)	μ	动态粘度 ($Pa\cdot s$)
g	重力加速度 ($米/秒^2$)	ξ	质量分数 (–)
h	焓 ($焦耳\ 千克^{-1}$)	ρ	密度 ($kg\ m^{-3}$)
$\bar{h}$	部分质量焓 ($焦耳\ 千克^{-1}$)	下标	
$\dot{m}$	质量通量 ($kg\ m^{-2}s^{-1}$)	o	入口条件
p	压力 (Pa)	abs	吸收
$\dot{q}$	热通量 (Wm^{-2})	avg	平均
T	温度 (K)	A	吸收质, 例如液态水
u	纵向速度 ($米/秒^1$)	des	解吸
x	纵向坐标, 流向 (m)	eq	平衡
y	横向坐标 (m)	if	界面
希腊字母		s	吸收剂溶液, 例如LiBr – H ₂ O
$\dot{\Gamma}$	单位长度质量流量 ($千克\ 米^{-1}秒^{-1}$)	v	蒸汽, 例如水蒸气
Δh_{sor}	吸附热 ($焦耳\ 千克^{-1}$)	w	Wall

废热源 (Meyer等人, 2009)。最近, Bilir等人 (2015) 研究了以引射器作为膨胀级的蒸气制冷循环的行为。他们对整个系统的火用分析表明, 喷射器膨胀制冷循环的总性能系数比常规制冷循环高出7–12%。因此, 从经济、运行和环境角度来看, 其应用具有吸引力, 用于替代或支持常规的压缩-膨胀装置 (Sumeru等人, 2012)。喷射器还可用于利用低压储罐、工业真空生成以及呼吸器内的气体混合。读者可参考Chunnanond和Aphornratana (2004), Elbel和Hrnjak (2008) 以及Elbel(2011) 的综述, 以详细了解喷射器系统在制冷或空调应用中的情况。

过去几十年间, 发表了大量关于各种制冷剂、不同喷射器设计以及单相和两相流的实验或理论工作。关于超音速喷射器的文献过于丰富, 难以对该主题进行全面综述。读者可参考Milazzo等人 (2014) 的综述, 以了解喷射器制冷理论和实验工作的最新概况。以下, 重点聚焦于单相超音速喷射器。

鉴于引射器尺寸较小且需要隔热, 实验研究常局限于全局测量: 质量流量、入口和出口的温度与压力, 或最近在混合室沿程进行的壁面压力测量 (Mazzelli和Milazzo, 2015)。Bartosiewicz等人 (2005) 曾尝试通过侵入式压力探针测量喷射器中心线的压力。仅有少数工作采用局部方法聚焦于激波结构。Marinovski等人 (2009) 通过激光层析成像研究了湿空气喷射器中发生的液滴凝结现象, 并揭示了激波串结构。Bouhanguel和Desevaux (2011) 扩展了这项工作, 使用不同的照明光源、偏振方向和示踪剂类型来研究超音速

空气喷射器的特定流动特征。Zhu和Jiang (2014) 通过光学纹影测量研究了两类喷射器内部的激波结构。他们建立了激波波长对一次流入口与出口压力比的依赖关系。还表明, 当喷射器达到设计工况时出现斜激波, 而非设计工况时则出现正激波。

数值模拟成为研究超音速喷射器内部局部流动特征的有力工具。由于喷射器几何结构相对复杂, 且涉及非常复杂的流动和能量传递现象 (激波、薄边界层、可压缩性效应、相变等), 所有作者都通过使用商业软件的二维轴对称模拟来关注喷射器的物理特性。Bartosiewicz等人 (2005年) 评估了六种双方程湍流模型在单相超音速喷射器中的性能。k–ε RNG和SST模型与实验数据相比整体吻合最佳。Zhu和Jiang (2014年) 部分证实了这一点, 他们表明k–ε RNG模型在激波预测方面比realizable和标准k–ε以及SST模型更优。他们建议对近壁区域进行详细建模, 以准确预测喷射器内部的流动结构。Scott等人 (2008年, 2011年) 使用Phoenix软件对以R245fa为工质的超音速喷射器进行了大量计算。针对不同几何结构和操作条件, 他们在引射比方面获得了与实验数据非常好的吻合, 最大偏差为10.8%。Cai和He (2013年) 研究了不同气体模型、湍流模型和几何变化对蒸汽超音速喷射器预测结果的影响。理想气体模型将引射比低估了20–40%。最近, Mazzelli和Milazzo (2015年) 对使用R245fa的超音速喷射器制冷机进行了实验和数值联合分析。结果表明, 壁面粗糙度对设计工况下的引射比影响微弱, 但会显著降低临界背压。全局喷射器性能仅轻微依赖于所选气体模型。Pianthong等人 (2007年) 和

Sriveerakul et al. (2007) were the first to perform 3D numerical simulations for a supersonic steam ejector coming to the same conclusion that there is no apparent 3D effect. Therefore 2D axisymmetric calculations may be sufficient to investigate such flow configurations. Sriveerakul et al. (2007) used a $k-\epsilon$ realizable turbulence model, the perfect gas equation of state and a coupled-implicit density based solver. Though having obtained a good overall agreement with their own experimental data, the authors suggest the use of real gas models and wall heat transfer. Their work was extended by Ruangtrakoon et al. (2013) to quantify the effect of primary nozzle geometries over the local flow features inside the ejector.

All these research groups did not reach a real consensus about the CFD (computational fluid dynamics) model, which could be confidently used to investigate the fluid flow and heat transfer in various types of supersonic single-phase ejectors. Moreover, no particular attention was paid to the influence of the solver or the numerical schemes on the predicted results. The purpose of this work is then to carefully build a CFD model offering the best compromise between accuracy and computational efforts to investigate in details the local flow features of a supersonic single-phase ejector working with R134a. The experimental database established by García del Valle et al. (2014) will be used for validation purposes.

The paper is organized as follows. The numerical modeling including the flow parameters, the numerical solver and the thermodynamic and turbulence models are presented in Section 2. Numerical benchmarks of three thermodynamic models and four two-equation turbulence models are performed in Section 3 before some concluding remarks in Section 4. The influence of the outlet temperature on the ejector performance and local flow features and the corresponding exergy analysis among other things will be explained in Part 2 (Croquer et al., 2015).

2. Numerical modeling

Two-dimensional axisymmetric and steady state calculations have been performed using the commercial software ANSYS Fluent v.15 for a single-phase supersonic ejector with R134a as the working fluid. A numerical benchmark of various high- and low-Reynolds number approaches and gas models is achieved considering the ejector recently designed by García del Valle et al. (2014).

Table 1 – Main dimensions of the ejector.	
Parameter	Value [mm]
Primary nozzle throat diameter, n_d	2.00
Primary nozzle exit diameter, d	3.00
Nozzle exit position, NXP	–5.38
Mixing chamber diameter, D	4.80
Mixing chamber length, l	41.39
Diffuser length, L	120.15
Diffuser exit diameter, e_d	20

2.1. Geometrical modeling

The geometry and operating conditions are based on the experimental set-up developed by García del Valle et al. (2014). They carried out an experimental analysis of three supersonic ejectors (models A, B and C) with refrigerant R134a. The differences between these three models are both the position of the primary nozzle and the shape of the mixing chamber. Their ejector with the geometry A operates over a wider range of boiler saturation temperatures compared to geometries B and C, while guaranteeing comparable performance. Geometry A has then been considered in this work. Figure 1 shows a schematic view of the geometry, while its main dimensions are summarized in Table 1. The length of the primary nozzle is fixed by the other parameters.

The area ratio A , defined as the ratio between the mixing chamber section and the primary nozzle section, is $A = (D/n_d)^2 = 5.76$ in the present work. According to Huang et al. (1999), higher area ratios induce higher values of the entrainment ratio ω with some optimal values around $A = [7-16]$ depending on the flow configurations and refrigerant. The length of the diffuser is here equal to $L = 120.15 \text{ mm} \approx 25D$, far from the value $L = 8D$ recommended by Henzler (1983). The present ejector appears to have not been optimally designed. Its other geometrical characteristics are fully displayed in García del Valle et al. (2014).

The geometry of the secondary inlet section is always quite complex in real systems and generally not fully characterized in most of the experimental works. In the experiments of García del Valle et al. (2014), the only available information is that the secondary inlet has a classical shape with a 30° half-angle conical entry. Moreover, as the present simulations assume the flow as being 2D axisymmetric, some assumptions are required to model this part of the ejector. A sensitivity analysis of the secondary inlet geometry has then been carried out to rule out any major

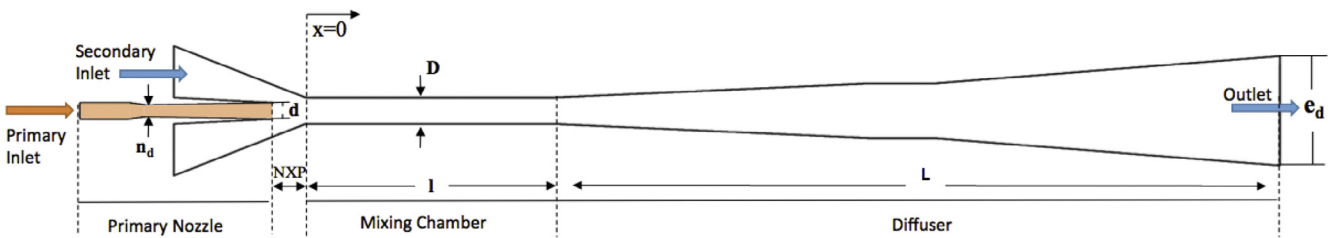


Fig. 1 – Schematic view of the ejector with relevant notations.

Sriveerakul等人 (2007年) 首次对超音速蒸汽喷射器进行了三维数值模拟，得出相同结论：没有明显的三维效应。因此，二维轴对称计算可能足以研究此类流动配置。Sriveerakul等人 (2007年) 使用了 $k-\epsilon$ realizable 湍流模型、理想气体状态方程和耦合隐式密度基求解器。尽管与自身实验数据获得了良好的整体一致性，但作者建议使用真实气体模型和壁面传热。他们的工作由 Ruangtrakoon 等人 (2013年) 延伸，以量化主喷嘴几何形状对喷射器内部局部流动特征的影响。

所有这些研究团队对于能够自信地用于研究各种类型超音速单相喷射器中流体流动和传热的计算流体动力学 (CFD) 模型并未达成真正共识。此外，没有特别关注求解器或数值格式对预测结果的影响。因此，本工作的目的是仔细构建一个在精度和计算工作量之间提供最佳折衷的CFD模型，以详细研究使用R134a的超音速单相喷射器的局部流动特征。由García del Valle等 (2014) 建立的实验数据库将用于验证。

本文结构如下。包括流动参数、数值求解器以及热力学和湍流模型的数值建模将在第2节中介绍。在第3节中对三个热力学模型和四个双方程湍流模型进行数值基准测试，然后在第4节中给出一些结论。出口温度对喷射器性能和局部流动特征的影响以及相应的火用分析等内容将在第二部分 (Croquer等, 2015) 中解释。

2. 数值建模

使用商业软件 ANSYS Fluent v.15 对以 R134a 为工作流体的单相超音速喷射器进行了二维轴对称稳态计算。针对 García del Valle 等人 (2014) 最近设计的喷射器，实现了多种高雷诺数和低雷诺数方法及气体模型的数值基准。García del Valle 等人 (2014)

表1 – 引射器的主要尺寸。	
参数	数值 [mm]
主喷嘴喉部直径, n_d	2.00
主喷嘴出口直径, d	3.00
喷嘴出口位置, NXP	–5.38
混合室直径, D	4.80
混合室长度, l	41.39
扩压器长度, L	120.15
扩压器出口直径, e_d	20

2.1. 几何建模

几何构型和操作条件基于 García del Valle 等人 (2014) 开发的实验装置。García del Valle 等人 (2014) 对三种超音速喷射器（模型 A、B 和 C）进行了实验分析，使用制冷剂 R134a。这三种模型之间的差异在于主喷嘴的位置和混合室的形状。与几何构型 B 和 C 相比，几何构型 A 的喷射器在更宽的锅炉饱和温度范围内运行，同时保证了可比的性能。因此，本文考虑了几何构型 A。图 1 显示了几何构型的示意图，其主要尺寸总结于表 1。主喷嘴的长度由其他参数确定。

面积比 A ，定义为混合室截面与主喷嘴截面之比，为 $A = (D/n_d)^2 = 5.76$ 。在本文中。根据 Huang 等人 (1999)，较高的面积比会带来较高的引射比 ω ，且存在一些最优值在 $A = [7-16]$ 之间，取决于流动构型和制冷剂。扩压器长度在此等于 $L = 120.15 \text{ mm} \approx 25D$ ，远低于 $L = 8D$ 所推荐的值 (Henzler, 1983)。当前引射器似乎未经过优化设计。其其他几何特征在 García del Valle 等人 (2014) 中有详细展示。

在实际系统中，二次进口段的几何形状通常非常复杂，并且大多数实验研究中并未完全描述其特征。在García del Valle等 (2014) 的实验中，唯一可用的信息是二次进口具有经典形状，带有 30° 半角锥形入口。此外，由于本次模拟假设流动为二维轴对称，因此需要对引射器的这一部分进行一些假设。随后，对二次进口几何形状进行了敏感性分析，以排除任何对引射器整体性能的重大影响。

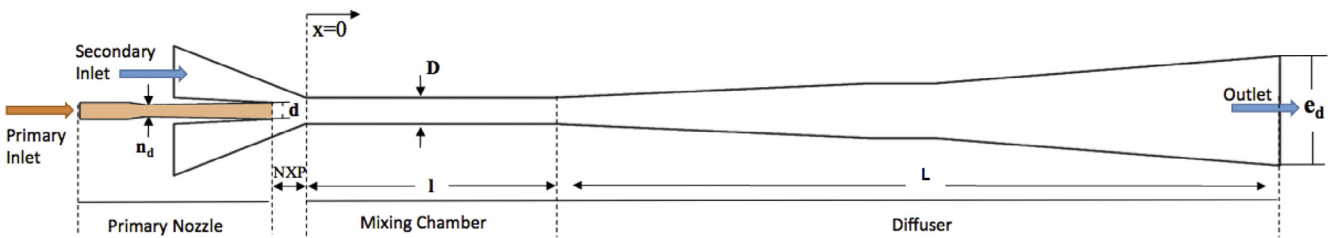


图1 – 引射器示意图及相关符号。

influence over the global performance of the ejector. Five different cases have been considered by slightly varying the inclination angles of the secondary inlet walls and the passage section of the secondary fluid. Similar results for the entrainment ratio (maximum deviation of 2.3%) have been obtained for the five distinct geometries (see in Croquer et al., 2015a). In the following, the geometry A (resp. B) for the secondary inlet, as defined by Croquer et al. (2015a), has been considered in low- (resp. high-) Reynolds number calculations.

2.2. Flow solver

The governing equations (conservations of mass, momentum and total energy) for a compressible fluid are solved using the commercial software ANSYS Fluent v.15, based on the finite volume method. The flow is assumed as being 2D axisymmetric. To our knowledge, there is indeed no evidence reported in the literature of three-dimensional effects in supersonic ejectors (Pianthong et al., 2007). Unsteady calculations have been first performed but all leading to steady-state solutions. The flow has been then considered in a steady-state regime.

A second-order upwind scheme is used to discretize the advective terms of each equation, except for the pressure equation. For this equation, the PRESTO! scheme designed for flows involving steep pressure gradients has been chosen. The diffusive terms are discretized using second-order central differenced schemes. Gradients are evaluated by a least-squares approach.

After the discretization process, a system of algebraic equations is obtained and may be solved using either a density- or pressure-based algorithm. From a historical point of view, a density-based solver is usually required to solve supersonic flows in the presence of shock waves. Recently, pressure-based solvers based on the SIMPLE algorithm or any of its derivatives have nevertheless demonstrated their capability to deal with highly compressible flows with shock waves in single- (Yadzani et al., 2012; Zhu and Jiang, 2014) or two-phase (Li and Li, 2011) flow ejectors. In the present case, the pressure-based algorithm Coupled, with full pressure–velocity coupling, has been used for all calculations. The Coupled algorithm solves the momentum and pressure-based continuity equations simultaneously. The full implicit coupling is achieved through an implicit discretization of pressure gradient terms in the momentum equations, and an implicit discretization of the face mass flux, including the Rhie–Chow pressure dissipation terms (ANSYS FLUENT Theory Guide Release, 2013). The energy equation is solved in a second step and density is computed through an equation of state $\rho = f(P, T)$ (see Section 2.3). For this particular application, the Coupled algorithm appeared to be much more stable than any available density-based solvers. A high-order term relaxation technique is also applied throughout the entire computation to ensure convergence smoothness.

2.3. Thermodynamic model

Three different equations of state are considered to evaluate the fluid density ρ of the R134a refrigerant as a function of the thermodynamic variables, pressure P and temperature T :

- The Perfect Gas model. Thermal and hydrodynamic fluid properties are assumed to be constant, apart from the

density, which obeys the ideal gas equation. Due to its simplicity, this model is still widely used by different authors to compute supersonic flows in ejectors (Bartosiewicz et al., 2005; Pianthong et al., 2007; Sriveerakul et al., 2007; Zhu and Jiang, 2014).

- The Redlich–Kwong–Soave (RKS) equation of state. The RKS equation is a pressure explicit cubic three-parameter equation of state (1), known for its simplicity and accuracy (Camporese et al., 1985). It requires the knowledge of only the critical pressure P_c , critical temperature T_c , acentric factor f and molar mass M of the fluid:

$$P = \frac{RT}{v-b} - \frac{a(T)}{v(v+b)} \quad (1)$$

$$a(T) = 0.42748 \frac{R^2 T_c^2}{P_c} \alpha(T) \quad (2)$$

$$b = 0.08664 \frac{RT_c}{P_c} \quad (3)$$

$$\alpha(T) = \left[1 + (0.480 + 1.574f - 0.176f^2) \left(1 - \left(\frac{T}{T_c} \right)^{1/2} \right) \right]^2 \quad (4)$$

- The REFPROP 7.0 equation database. The REFPROP model (Lemmon et al., 2013) for refrigerant R134a is based on the formulation of Tillner-Roth and Baehr (1994). Fluid thermodynamic and transport properties (viscosity, specific heat, and thermal conductivity) are computed using the Helmholtz free energy equation of state. This equation depends on 21 parameters, obtained from statistical analysis and least-squares fitting of the most accurate R134a measurements available at that time (Tillner-Roth and Baehr, 1994). This model accurately represents real gas behavior in the temperature range [170–455] K and for pressures up to 70 MPa.

For the perfect gas model, all thermo-physical properties (viscosity, specific heat, and thermal conductivity) are considered as constant and fixed to the average between the inlet and outlet values. The same procedure has been applied for the RKS model in terms of viscosity and thermal conductivity. The specific heat capacity is related to differentials of pressure according to temperature and density through the RKS equation of state. For all models and whatever the operating conditions (see Table 2), the fluid is assumed to remain in the gas phase. Its properties are summed up in Table 3.

2.4. Turbulence modeling

Various two-equation turbulence models available within Fluent v.15 have been compared to the experimental data of García del Valle et al. (2014) in their high- or low-Reynolds number formulation. In their high-Reynolds number form, the following four two-equation turbulence models have been considered:

- A standard $k-\epsilon$ model, known for its robustness, economy, and reasonable accuracy over a wide range of turbulent

通过略微改变二次入口壁面的倾斜角度和二次流体的流通截面, 考虑了五种不同情况。对于这五种不同的几何构型, 获得了相似引射比的结果 (最大偏差为2.3%) (见Croquer等, 2015a)。在下文中, 采用由Croquer等 (2015a) 定义的二次进口几何构型A (对应B) 在低 (对应高) 雷诺数计算中。

2.2. 流场求解器

使用基于有限体积法的商业软件 ANSYS Fluent v.15 求解可压缩流体的控制方程 (质量、动量和总能量守恒)。流动假设为二维轴对称。据我们所知, 文献中确实没有关于超音速喷射器中三维效应的报道 (Pianthong 等人, 2007)。首先进行了非定常计算, 但所有计算均得到了稳态解。因此, 流动被认为处于稳态工况。

除压力方程外, 每个方程的对流项均采用二阶迎风格式离散。对于压力方程, 选择了专为涉及陡峭压力梯度的流动设计的 PRESTO! 格式。扩散项采用二阶中心差分格式离散。梯度通过最小二乘法评估。

离散化过程后, 得到代数方程组, 可通过密度基或压力基算法求解。从历史角度来看, 密度基求解器通常用于求解存在激波的超音速流动。然而近年来, 基于 SIMPLE 算法或其衍生算法的压力基求解器已证明能够处理单相 (Yadzani 等人, 2012; Zhu 和 Jiang, 2014) 或两相 (Li 和 Li, 2011) 喷射器中带有激波的高度可压缩流动。在本例中, 所有计算均采用全压力-速度耦合的压力基 Coupled 算法。Coupled 算法同时求解动量和基于压力的连续性方程。全隐式耦合通过动量方程中压力梯度项的隐式离散以及面质量通量的隐式离散 (包括 Rhie-Chow 压力耗散项) 来实现 (ANSYS FLUENT 理论指南发布, 2013)。能量方程在第二步求解, 密度通过状态方程 $\rho = f(P, T)$ 计算 (见第 2.3 节)。对于此特定应用, Coupled 算法比任何可用的密度基求解器都稳定得多。在整个计算过程中还应用了高阶项松弛技术, 以确保收敛平滑。

2.3. 热力学模型

为评估R134a制冷剂流体密度 ρ 随热力学变量 (压力 P 和温度 T) 的变化, 考虑了三种不同的状态方程:

- 理想气体模型。除密度外, 假设热力和流体动力学流体特性是恒定的, 密度遵循理想气体方程。由于其简单性, 该模型仍被不同作者广泛用于计算引射器中的超音速流动 (

密度, 它遵循理想气体方程。由于其简单性, 该模型仍被不同作者广泛用于计算引射器中的超音速流动 (Bartosiewicz等人, 2005; Pianthong等人, 2007; Sriveerakul等人, 2007; Zhu和Jiang, 2014)。

- Redlich–Kwong–Soave (RKS) 状态方程。RKS方程是一个压力显式的立方三步状态方程 (1), 以其简单性和准确性而闻名 (Camporese等, 1985)。它只需要了解流体的临界压力 P_c 、临界温度 T_c 、偏心因子 f 和摩尔质量 M :

$$P = \frac{RT}{v-b} - \frac{a(T)}{v(v+b)} \quad (1)$$

$$a(T) = 0.42748 \frac{R^2 T_c^2}{P_c} \alpha(T) \quad (2)$$

$$b = 0.08664 \frac{RT_c}{P_c} \quad (3)$$

$$\alpha(T) = \left[1 + (0.480 + 1.574f - 0.176f^2) \left(1 - \left(\frac{T}{T_c} \right)^{1/2} \right) \right]^2 \quad (4)$$

- REFPROP 7.0 方程数据库。REFPROP 模型 (Lemmon等, 2013) 针对制冷剂R134a, 基于 Tillner-Roth 和 Baehr (1994) 的公式。流体热力学和输运性质 (粘度、比热和导热系数) 使用亥姆霍兹自由能状态方程计算。该方程依赖于21个参数, 这些参数通过对当时最准确的R134a测量数据进行统计分析和最小二乘拟合获得 (Tillner-Roth 和 Baehr, 1994)。该模型在温度范围 [170–455] K 和压力高达70 MPa 下准确表示真实气体行为。

对于理想气体模型, 所有热物理性质 (粘度、比热和导热系数) 均视为常数, 并固定为进口与出口值的平均值。对于RKS模型, 在粘度和导热系数方面采用了相同的处理方式。比热容通过RKS状态方程与压力和温度、密度的微分相关。对于所有模型及任何操作条件 (见表2), 假设流体始终处于气相。其性质总结于表3。

2.4. 湍流建模

在Fluent v.15中可用的各种双方程湍流模型, 以其高雷诺数或低雷诺数公式, 与García del Valle等人 (2014) 的实验数据进行了比较。在高雷诺数形式中, 考虑了以下四种双方程湍流模型:

- 标准 $k-\epsilon$ 模型以其在广泛湍流范围内的鲁棒性、经济性和合理的精度而闻名。

Table 2 – Operating conditions simulated in the present work and corresponding to the experiments of García del Valle et al. (2014) .								
Operation	Primary inlet (boiler)		Secondary inlet (evaporator)		Outlet (condenser)		Ejector performance	
Point (OP)	P [kPa]	T _{sat} [°C]	P [kPa]	T _{sat} [°C]	P [kPa]	T _{sat} [°C]	ω_{exp} [–]	p^{ratio} [–]
1	2598.04	79.37	414.608	10.00	757.222	29.41	0.494	1.826
2	2888.8	84.39	414.608	10.00	826.573	32.48	0.398	1.994
3	3188.14	89.15	414.608	10.00	897.115	35.41	0.339	2.164

Table 3 – Thermo-physical properties obtained by the three thermodynamic models for OP2 (total pressure P ∈ [356.43–2950.56] kPa and total temperature T ∈ [16.34–95.4] °C).			
Properties/model	Perfect gas	RKS	REFPROP 7.0
kinematic viscosity (×10 ^{–5} Pa.s)	1.44	1.44	1.05–1.62
Prandtl number (–)	1.54	1.55–1.83	0.78–1.12
Thermal conductivity (W/m/K)	0.021	0.021	0.011–0.023
Heat capacity (J/kg/K)	2240.1	2255.5–2667.7	871.8–1590
Density (kg/m³)	8.64–93.03	9.4–139.55	11.72–144.5

- flows. The assumption is that the flow is fully turbulent and the effects of viscosity are negligible.
- A $k-\epsilon$ RNG model. This model is derived from the instantaneous Navier–Stokes equations, using a “renormalization group” (RNG) method. The analytical derivation results in a model with constants different from those in the standard $k-\epsilon$ model, and additional terms and functions in the transport equations for the turbulence kinetic energy k and its dissipation rate ϵ .
 - A $k-\epsilon$ realizable model. This model is an improved version of the first two ones. The turbulence viscosity is evaluated in a different way and a new transport equation for the dissipation rate ϵ has been derived from an exact equation for the transport of the mean-square vorticity fluctuation taking into account some mathematical constraints. It is expected that this model is able to predict a more accurate spreading rate of the jet ([ANSYS FLUENT Theory Guide Release, 2013](#)) at the outlet of the primary nozzle.
 - A shear–stress transport (SST) $k-\omega$ model. The SST model is a hybrid model, which blends the accurate formulation of the standard $k-\omega$ model in the near-wall region with the free-stream independence of the $k-\epsilon$ model in the far field. Compared to the standard $k-\omega$ model, it incorporates also a damped cross-diffusion derivative term in the transport equation of the specific dissipation rate ω and the turbulence viscosity is modified to account for the transport of the turbulent shear stress.

The flow in the near-wall regions is not directly computed by such approaches but rather approximated using appropriate wall functions, which relax the constraints in terms of required mesh grids in these regions and then considerably reduces the computational time. All these two-equation models in their high-Reynolds number formulation offer a relatively good compromise between accuracy and computational cost

even if none of them is capable of having good performance for all possible flow configurations. It appears then necessary to perform a numerical benchmark of these models for the specific ejector system designed by [García del Valle et al. \(2014\)](#). For the low-Reynolds number approach, only the $k-\omega$ SST model has been considered in the following. The reasons are twofold: some previous numerical studies ([Bartosiewicz et al., 2005](#); [Zhu and Jiang, 2014](#)) showed its overall superiority over low-Reynolds number $k-\epsilon$ models and computations using the standard $k-\epsilon$ model in its low-Reynolds number formulation were found here to be more unstable for this range of operating conditions.

2.5. Numerical settings

Three operating conditions corresponding to the experiments of [García del Valle et al. \(2014\)](#) have been considered in the present work. At both inlets and the outlet, the pressure is prescribed based on the saturation temperature, denoted T_{sat} . The compression ratio p^{ratio} , defined as the ratio of static pressures between the outlet and the secondary inlet, is then an input in the present simulations. All values for the saturation temperatures and static pressures imposed at the inlets and the outlet are displayed in [Table 2](#) together with the experimental values of the entrainment ratio ω_{exp} and the computed compression ratio p^{ratio} . It is noticeable that inlet temperatures include an overheat of 10°C (e.g. the primary fluid temperature for OP 1 is fixed to 89.37°C at the inlet). These working conditions are particularly appropriate and suitable for refrigeration systems. The low boiler temperatures enable heat utilization from various cheap sources such as industrial waste heat or renewable energy sources, at the same time the low evaporator temperatures enable to meet the refrigerating or air-conditioning requirement (heat is rejected by the compression cycle) while guaranteeing a good performance of the system. It guarantees also the absence of condensation phenomena, allowing for the adoption of a more simpler gas model. One recalls that the flow is assumed to be steady-state and 2D axisymmetric, neglecting three-dimensional effects ([Pianthong et al., 2007](#)). Compared to the average velocities reached inside the ejector, velocities at the inlets and the outlet are almost negligible. Thus, the total values for pressure and temperature are supposed to be equal to their static values, which is a usual approximation in CFD analysis of supersonic ejectors. Turbulence intensities at the inlets are set at 5%. It has been checked that considering other turbulence levels leads to indistinguishable results. Walls are considered as adiabatic and hydrodynamically smooth surfaces. [Mazzelli and Milazzo \(2015\)](#) showed that

表2 – 本研究模拟的操作条件，对应于 García del Valle等人（2014） 的实验。								
操作	主入口 (锅炉)		二次进口 (蒸发器)		出口 (冷凝器)		引射器性能	
工作点 (OP)	P [千帕]	T _{sat} [°C]	P [千帕]	T _{sat} [°C]	P [千帕]	T _{sat} [°C]	ω_{exp} [–]	p^{ratio} [–]
1	2598.04	79.37	414.608	10.00	757.222	29.41	0.494	1.826
2	2888.8	84.39	414.608	10.00	826.573	32.48	0.398	1.994
3	3188.14	89.15	414.608	10.00	897.115	35.41	0.339	2.164

表3 – 三种热力学模型获得的OP2热物理性质（总压力三种热力学模型获得的OP2（总压力 P ∈ [356.43–2950.56] . . kPa, 总温度 T ∈ [16.34–95.4] °. . C）。			
性质/模型	完美 gas	RKS	REFPROP 7.0
运动黏度 (×10 ^{–5} Pa.s)	1.44	1.44	1.05–1.62 .
普朗特数 (–)	1.54	1.55–1.83 .	0.78–1.12 .
导热系数 (W/m/K)	0.021	0.021	0.011–0.023 .
热容 (J/kg/K)	2240.1	2255.5–2667.7 .	871.8–1590.
密度 (kg/m³)	8.64–93.03 .	9.4–139.55 .	11.72–144.5 .

该流动假设是完全湍流，且黏性效应可忽略不计。

- A $k-\epsilon$ RNG模型。该模型是从瞬态纳维-斯托克斯方程出发，使用“重整化群”（RNG）方法推导得出的。解析推导得到的模型具有与标准 $k-\epsilon$ 模型不同的常数，以及湍动能 k 及其耗散率 ϵ 输运方程中的附加项和函数。
- A $k-\epsilon$ 可实现模型。该模型是前两种模型的改进版本。湍流粘度的评估方式不同，并且耗散率的新的输运方程 ϵ 是从均方涡量脉动输运的精确方程推导而来，同时考虑了某些数学约束。预计该模型能够预测主喷嘴出口处射流更准确的扩展速率（[ANSYS FLUENT理论指南, 2013版](#)）。
- A剪切应力输运（SST） $k-\omega$ 模型。SST模型是一种混合模型，它将标准 $k-\omega$ 模型在近壁区域的准确公式与 $k-\epsilon$ 模型在远场的自由流独立性相结合。与标准 $k-\omega$ 模型相比，它还在比耗散率 ω 的输运方程中加入了阻尼交叉扩散导数项，并修改了湍流粘度以考虑湍流剪应力的输运。

近壁区域内的流动并非由这些方法直接计算，而是使用适当的壁面函数进行近似，这放宽了这些区域网格数量的限制，从而大幅减少计算时间。所有采用高雷诺数公式的这些双方程模型在精度和计算成本之间提供了相对良好的折中。

即使没有一种模型能在所有可能的流动配置中表现良好，因此有必要对这些模型进行数值基准测试，针对[García del Valle等人（2014）](#)设计的特定引射器系统。

对于低雷诺数方法，在后续中仅考虑了 $k-\omega$ SST模型。原因有二：一些先前的数值研究（[Bartosiewicz等人, 2005](#)；[Zhu和Jiang, 2014](#)）表明，相比其他低雷诺数 $k-\epsilon$ 模型，该模型具有整体优越性，并且在此操作条件范围内，使用标准 $k-\epsilon$ 模型及其低雷诺数公式的计算被发现更加不稳定。

2.5. 数值设置

本工作考虑了与[García del Valle等人（2014）](#)实验相对应的三种操作条件。在入口和出口处，基于饱和温度（记为 T_{sat} ）设定压力。压缩比 p^{ratio} 定义为出口静压与二次进口静压之比，在当前模拟中作为输入参数。所有入口和出口处施加的饱和温度和静压值列于[表2](#)，同时给出实验引射比 ω_{exp} 和计算压缩比 p^{ratio} 。值得注意的是，入口温度包含10°C的过热度（例如，OP 1的一次流体温度在入口处设为89.37°C）。这些工作条件特别适合制冷系统。低锅炉温度使得可以利用来自各种廉价来源（如工业废热或可再生能源）的热量，同时低蒸发器温度能够满足制冷或空调需求（热量通过压缩循环排出），同时保证系统良好性能。这也保证了无冷凝现象发生，从而可以采用更简单的气体模型。

需要指出的是，流动假设为稳态和二维轴对称，忽略三维效应（[Pianthong等人, 2007](#)）。与引射器内部达到的平均速度相比，入口和出口处的速度几乎可以忽略不计。因此，压力和温度的总值被认为等于其静态值，这是超音速引射器CFD分析中常见的近似。入口处的湍流强度设为5%。经检验，考虑其他湍流水平会得到难以区分的结果。

壁面被视为绝热且水力学光滑表面。[Mazzelli and Milazzo \(2015\)](#)表明

imposing an arithmetic roughness height equal to $3.5\text{ }\mu\text{m}$ improved the numerical results by slightly decreasing the entrainment ratio. Some numerical tests have been done here for two values of the roughness height: $2\text{ }\mu\text{m}$ and $20\text{ }\mu\text{m}$, keeping in mind that such quantity is not available in the experiments of [García del Valle et al. \(2014\)](#). Even for the lowest value, the predicted entrainment ratio drops far below the experimental values for OP2 and OP3 exhibiting all features of off-design conditions. For example, for OP3, the numerical predictions using the standard $k\text{--}\epsilon$ model and the REFPROP database give $\omega = 0.232$ for a roughness height equal to $2\text{ }\mu\text{m}$ and 0.028 for $20\text{ }\mu\text{m}$, which is to be compared to the experimental value $\omega_{\text{exp}} = 0.339$. All turbulence models provide similar results and confirm the results of [Mazzelli and Milazzo \(2015\)](#), who reported the same steepness and extension of the off-design region for similar condenser temperatures. The inner surface of the ejector designed by [García del Valle et al. \(2014\)](#) may be considered as a mirror-like surface and as a smooth wall in the CFD model.

A hybrid initialization based on boundary interpolation methods is used to start the calculations. A Laplace equation is solved to determine the velocity and pressure fields from the imposed boundary conditions. All other variables, such as temperature or turbulence quantities, are automatically patched based on domain averaged values.

A mesh sensitivity analysis was carried out for both low- and high-Reynolds number approaches to ensure the independence of the solution from the spatial discretization:

- For all turbulence models based on a high-Reynolds number formulation, the mesh grid was built using ANSYS Meshing software v.15 with standard wall functions for the near-wall treatment. An unstructured type mesh is used with 5 prismatic layers in the near-wall regions. A mesh grid with

51,000 cells has proven to be sufficient to get grid independent solutions ([Croquer et al., 2015a](#)). The maximum value for the wall coordinate remains then in the range (2010; 2013; [Camporese et al., 1985](#); [Croquer et al., 2015a, 2015b](#); [Henzler, 1983](#); [Huang et al., 1999](#); [Li and Li, 2011](#); [Ruangtrakoon et al., 2013](#); [Tillner-Roth and Baehr, 1994](#); [Yadzani et al., 2012](#)) depending on the RANS model, which is usually required when using wall functions.

- For the low-Reynolds number approach, the software ANSYS ICEM v.15 was used to build the structured mesh grid. It is composed of tetrahedral elements, which allows a greater control over the near-wall elements and their growing rate toward the free flow area. A mesh grid with 645,000 elements proves to be sufficient to get grid independent solutions. The wall coordinate remains less than 0.79, which ensures to have at least 18 mesh points to describe the linear and logarithmic regions of the velocity profile ([Croquer et al., 2015b](#)).

Details of the chosen grids are shown in [Fig. 2](#). In all cases, a mesh refinement in the radial direction is necessary in the mixing layer appearing at the trailing edge of the primary nozzle. As the mesh grids are sufficiently refined in the streamwise direction for both approaches, the adaptative mesh option does not improve the predictions of the CFD model and has been then switched off to reduce the calculation time.

Convergence is achieved when stable values of the static outlet temperature, total mass flow across the domain and residuals are under 10^{-4} for all conservation equations. It is typically reached shortly after 1000 iterations. All the computations have been performed using a workstation with 16 GB of RAM and a 4 core 3.40 GHz CPU. In average, the calculation time is around 1 hour and 3 hours for the high- and low-Reynolds number approaches respectively using 3 processors.

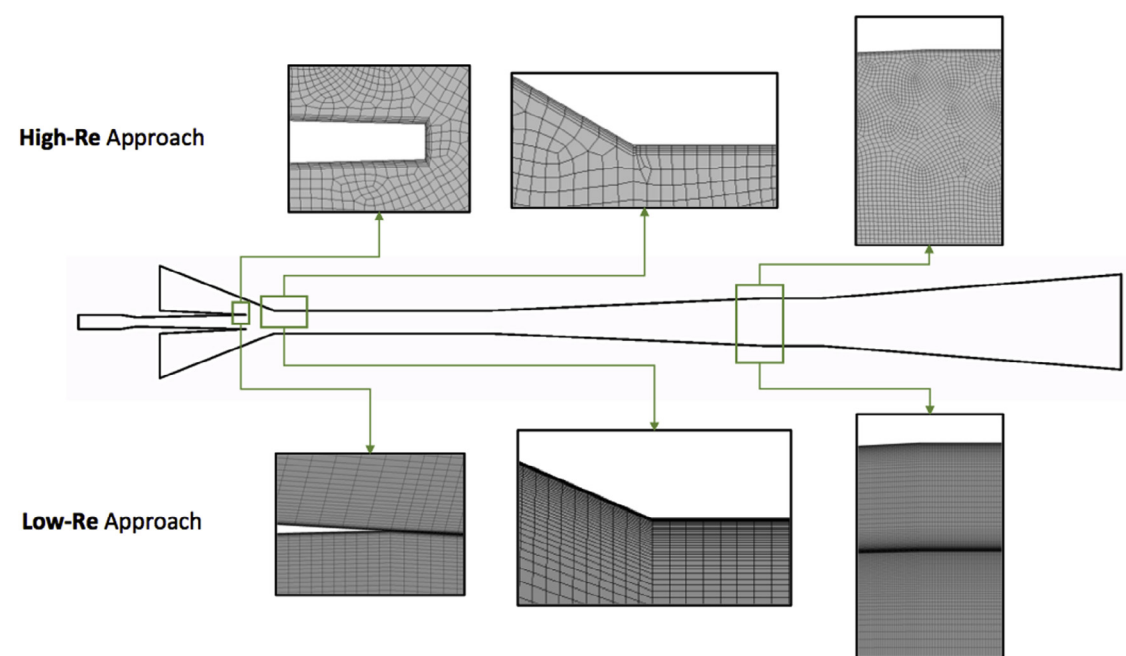


Fig. 2 – Details of the mesh grids used for the high- and low-Reynolds number turbulence models.

施加算术粗糙度高度为 $3.5\text{ }\mu\text{m}$ 可通过略微降低引射比来改善数值结果。本文针对两个粗糙度高度值进行了一些数值测试: $2\text{ }\mu\text{m}$ 和 $20\text{ }\mu\text{m}$, 需注意该量在 [García del Valle et al. \(2014\)](#) 的实验中不可得。即使对于最低值, 预测的引射比也远低于 OP2 和 OP3 的实验值, 展现出所有非设计工况的特征。例如, 对于 OP3, 使用标准 $k\text{--}\epsilon$ 模型和 REFPROP 数据库得出 $\omega = 0.232$ (粗糙度高度为 $2\text{ }\mu\text{m}$ 时) 和 0.028 (粗糙度高度为 $20\text{ }\mu\text{m}$ 时), 与实验值 $\omega_{\text{exp}} = 0.339$ 相比。所有湍流模型提供相似结果, 并确认了 [Mazzelli and Milazzo \(2015\)](#) 的结论, 他们报告了类似冷凝温度下非设计区域的相同陡度和延伸范围。 [García del Valle et al. \(2014\)](#) 设计的引射器内表面可视为镜面, 在 CFD 模型中作为光滑壁面处理。

采用基于边界插值方法的混合初始化来启动计算。通过求解拉普拉斯方程, 从施加的边界条件确定速度场和压力场。所有其他变量, 如温度或湍流量, 都基于域平均值自动进行赋值。

针对低雷诺数和高雷诺数方法进行了网格敏感性分析, 以确保解的空间离散化独立性:

- 对于所有基于高雷诺数公式的湍流模型, 网格是使用 ANSYS Meshing 软件 v.15 构建的, 并采用标准壁面函数处理近壁区域。使用非结构网格, 在近壁区域设置 5 层棱柱层。网格包含

51,000 个单元, 已被证明足以获得网格无关解 ([Croquer 等, 2015a](#))。壁面坐标的最大值保持在 (2010; 2013; [Camporese 等, 1985](#); [Croquer 等, 2015a, 2015b](#); [Henzler, 1983](#); [Huang 等, 1999](#); [Li 和 Li, 2011](#); [Ruangtrakoon 等, 2013](#); [Tillner-Roth 和 Baehr, 1994](#); [Yadzani 等, 2012](#)) 范围内, 具体取决于 RANS 模型, 这是使用壁面函数时的通常要求。

- 对于低雷诺数方法, 使用软件 ANSYS ICEM v.15 构建结构化网格。该网格由四面体单元组成, 能够更好地控制近壁面单元及其向自由流动区域的增长率。包含 645,000 个单元的网格足以获得网格无关解。壁面坐标保持在 0.79 以下, 确保至少有 18 个网格点来描述速度分布的线性和对数区域 ([Croquer 等, 2015b](#))。

所选网格的详细信息如图 2 所示。在所有情况下, 在主喷嘴尾缘出现的混合层中, 径向方向需要进行网格细化。由于两种方法的网格在流向上已充分细化, 自适应网格选项并未改善 CFD 模型的预测结果, 因此已关闭该选项以减少计算时间。

当出口静温、整个域的总质量流量和残差在所有守恒方程中稳定在以下时, 即达到收敛。通常在 1000 次迭代后不久即可实现。所有计算均在一台配备 16 GB 内存和 4 核 3.40 GHz 中央处理器的工作站上进行。平均而言, 高雷诺数方法和低雷诺数方法分别使用 3 个处理器时, 计算时间约为 1 小时和 3 小时。

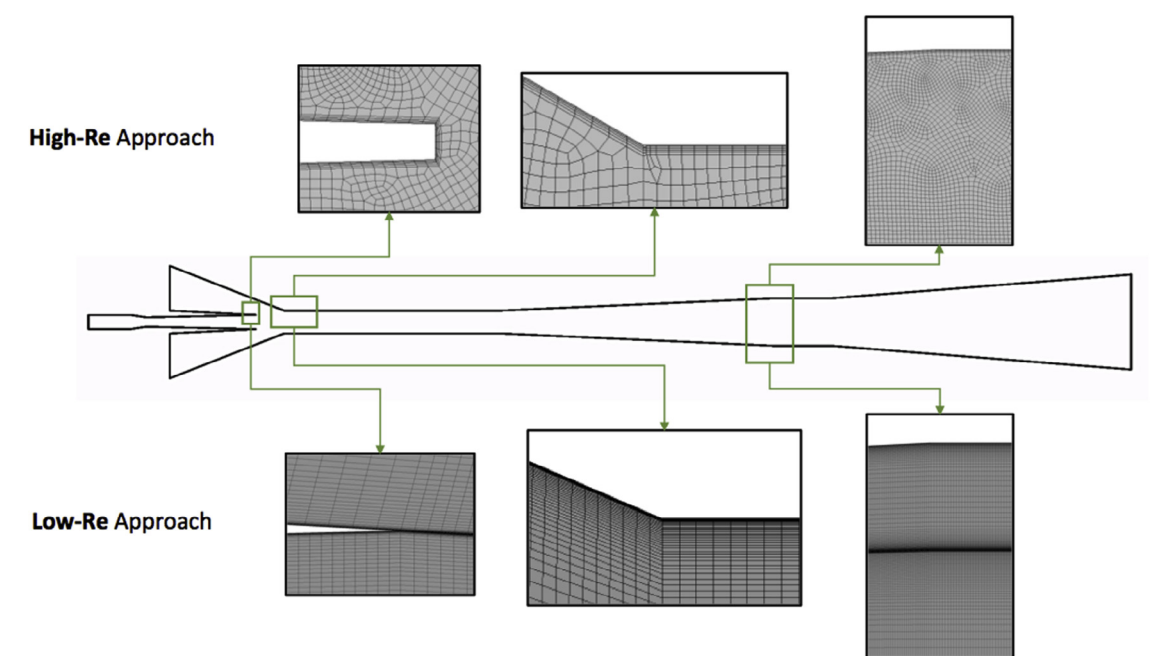


Fig. 2 – 用于高雷诺数和低雷诺数湍流模型的网格细节

3. Results and discussion

The results are compared to the experimental database of [García del Valle et al. \(2014\)](#) and discussed in terms of the entrainment ratio and the shock wave structure for three thermodynamic and four turbulence models.

3.1. Influence of the thermodynamic model

The present ejector uses the 1,1,1,2-tetrafluoroethane ($C_2H_2F_4$), also known as R134a or HFC-134a refrigerant. It is an environmentally benign fluid classified in the AHSRAE safety group A1 (not toxic and nonflammable). For these reasons, it has been gradually replacing more hazardous refrigerants like the HCFC22 in refrigeration or road transport systems. Its boiling point is -26.3°C such that R134a is always in gas phase for the present application. An equation of state is then required to evaluate its density $\rho(P,T)$, which strongly depends on local pressure P and temperature T . The calculations have been performed using the standard $k-\epsilon$ model in its high-Reynolds number formulation for three equations of state.

Figure 3 compares the three gas models in terms of the entrainment ratio ω for the three operating points. The perfect gas model fails to predict the good entrainment ratio whatever the operating conditions. The maximum deviation compared to the experimental values of [García del Valle et al. \(2014\)](#) is up to 19% at OP2 and OP3. The two real gas models provide essentially the same results with a very good agreement for OP1, the deviation with the experimental value being less than 1%. At OP 2 (resp. OP 3), deviations are 4.4% (resp. 5.8%) and 6% (resp. 3.4%) for the REFPROP 7.0 and RKS models respectively. The slight overprediction of the entrainment ratio ω increases for increasing values of the outlet pressure but the two real gas models show negligible differences among each other, such that no definitive conclusion can be drawn. Even if different ejectors, operating conditions and refrigerants have been considered, the present results do not support the conclusions of [García del Valle et al. \(2012\)](#). The authors showed indeed using an innovative 1D model based on a potential flow approach that real or ideal gas behaviors lead to very similar results in terms of entrainment ratio. It supports the conclusion that no universal 1D or CFD model may be recommended.

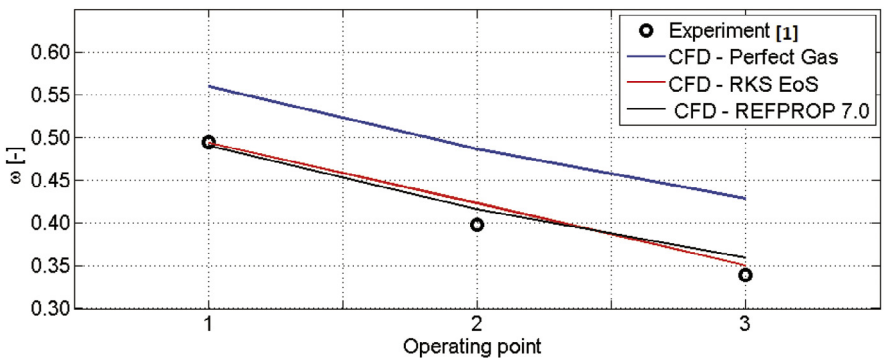


Fig. 3 – Entrainment ratio ω for the three operating points. Comparisons between three gas models using the standard $k-\epsilon$ model and the experimental data of [García del Valle et al. \(2014\)](#).

To go into more details, [Fig. 4a and b](#) displays the distributions of the static pressure and Mach number respectively in the streamwise direction at the centerline of the ejector for OP2. All three models provide essentially the same profiles with a normal shock wave of relatively weak intensity located around $x = 0.04\text{ m}$ at the outlet of the mixing chamber. As can be seen from the two zooms, the exact position of the shock wave slightly depends on the thermodynamic model. [García del Valle et al. \(2014\)](#) suggested that a system of shock waves appeared in the mixing area and it will be shown in [Section 3.2](#) that this discrepancy is more related to the choice of the turbulence model.

[Table 3](#) summarizes the main thermo-physical properties used or calculated by the three thermodynamic models. It shows in particular the importance of using real gas models as the heat capacity (and so the Prandtl number) may vary in a large range. Surprisingly, both real gas models predict very different results for the Prandtl number and the heat capacity, while predicting the same entrainment ratio and Mach and pressure distributions.

It may be of interest also to look at the distribution of the compressibility factor Z in this case. The compressibility factor is the ratio of the molar volume of a gas to the molar volume of an ideal gas at the same temperature and pressure and is evaluated through the equation of state. It may be seen as a useful indicator of the deviation from the perfect gas model. [Figure 5](#) displays the map of Z obtained by the model based on the REFPROP 7.0 database equation for OP2. The factor Z varies from 0.648 at the inlet of the primary nozzle to 0.935 in the constant section area where the shock wave appears. As expected, the deviation from the perfect gas law is then maximum at the inlet of the primary nozzle where the highest pressure levels are reached. Z is equal to 0.91 in the secondary fluid entrance region, close to the behavior of a perfect gas. The overprediction already evoked above in terms of the entrainment ratio is then related to inaccurate predictions of the heat and fluid flow inside the primary nozzle. Similar results have been obtained using the RKS equation of state with $0.672 \leq Z \leq 0.967$.

As a conclusion, the use of the perfect gas model to investigate supersonic ejectors working with R134a, though being more easy to use, is not advisable. The two real gas models perform quite well providing very similar results in terms of

3. 结果与讨论

结果与[García del Valle等人 \(2014\)](#)的实验数据库进行了比较, 并从引射比和激波结构的角度讨论了三种热力学模型和四种湍流模型。

3.1. 热力学模型的影响

本引射器使用1,1,1,2-四氟乙烷 ($C_2H_2F_4$), 也称为R134a或HFC-134a制冷剂。它是一种环境友好型流体, 属于AHSRAE安全组A1 (无毒且不可燃)。因此, 它已逐步取代制冷或道路运输系统中更具危害性的制冷剂, 如HCFC22。其沸点为 -26.3°C , 使得R134a在当前应用中始终处于气相。因此需要状态方程来评估其密度 $\rho(P,T)$, 该密度强烈依赖于局部压力 P 和温度 T 。计算采用标准 $k-\epsilon$ 模型的高雷诺数公式, 针对三种状态方程进行。

[图3](#)比较了三种气体模型在三个操作点的引射比 ω 。理想气体模型无法准确预测任何操作条件下的引射比。与[García del Valle等人 \(2014\)](#)的实验值相比, 在OP2和OP3时最大偏差高达19%。两种真实气体模型的结果基本相同, 在OP1时与实验值的偏差小于1%, 符合得非常好。在OP2 (或OP3) 处, REFPROP 7.0和RKS模型的偏差分别为4.4% (或5.8%) 和6% (或3.4%)。引射比的轻微高估 ω 随着出口压力的增加而增大, 但两种真实气体模型之间的差异可忽略不计, 因此无法得出明确结论。尽管考虑了不同的引射器、操作条件和制冷剂, 但当前结果并不支持[García del Valle等人 \(2012\)](#)的结论。作者通过一种基于势流方法的创新一维模型表明, 真实气体或理想气体行为在引射比方面导致非常相似的结果。这支持了没有通用的一维或CFD模型可以推荐的结论。

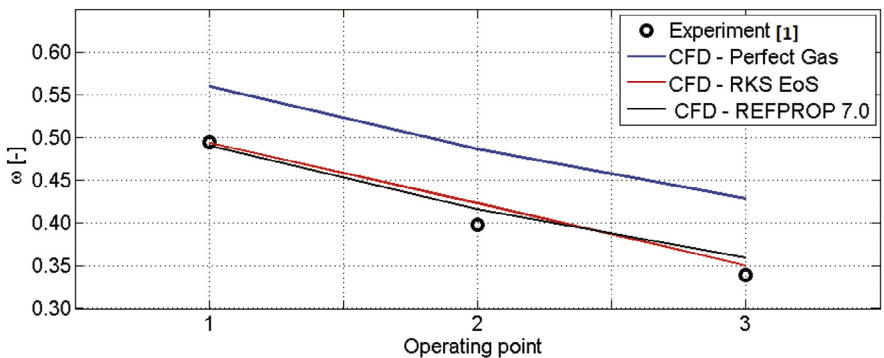


图 3 – 引射比 ω 对应三个运行点。使用标准 $k-\epsilon$ 模型和 [García del Valle 等人 \(2014\)](#) 的实验数据比较三种气体模型。

为了更详细地探讨, [图4a和b](#)分别显示了OP2时引射器中心线沿流向的静压和马赫数分布。三种模型都提供了基本相同的分布, 在混合室出口附近约 $x = 0.04\text{ m}$ 处出现强度相对较弱的正激波。从两个放大图可以看出, 激波的确切位置略微依赖于热力学模型。[García del Valle等人 \(2014\)](#)认为混合区域出现了一系列激波, 而将在第3.2节中说明这种差异更多与湍流模型的选择有关。

[表3](#)总结了三个热力学模型使用或计算的主要热物理性质。它特别显示了使用真实气体模型的重要性, 因为热容 (以及普朗特数) 可能在大范围内变化。令人惊讶的是, 两个真实气体模型在普朗特数和热容方面预测出截然不同的结果, 但却预测出相同的引射比以及马赫数和压力分布。

关注压缩因子 Z 在此案例中的分布也很有意义。压缩因子是气体摩尔体积与同温同压下理想气体摩尔体积的比值, 通过状态方程计算得出。它可以作为偏离理想气体模型的有用指标。[图5](#)展示了基于REFPROP 7.0数据库方程针对OP2的模型所获得的 Z 分布图。压缩因子 Z 从主喷嘴入口处的0.648变化到激波出现的等截面段处的0.935。正如预期, 与理想气体定律的偏差在主喷嘴入口处最大, 因为那里达到了最高压力水平。 Z 在二次流体入口区域等于0.91, 接近于理想气体的行为。前面提到的引射比高估与主喷嘴内部热量和流体流动预测不准确有关。使用RKS状态方程也获得了类似结果, $0.672 \leq Z \leq 0.967$ 。

总之, 使用理想气体模型研究以R134a为工质的超音速喷射器虽然更易用, 但并不可取。两种真实气体模型表现良好, 在喷射器性能和局部流动特征方面给出了非常相似的结果。

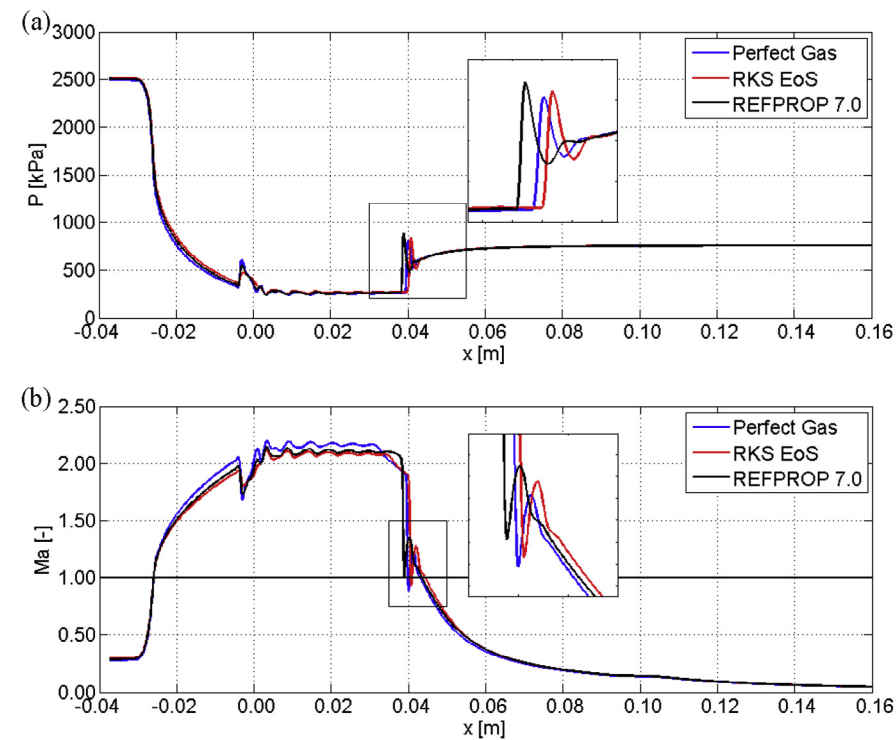


Fig. 4 – Comparisons of the gas models for OP2: (a) static pressure profile P (kPa) and (b) Mach number Ma distribution along the centerline of the ejector. Results obtained using the standard $k-\epsilon$ model.

ejector performance and local flow features. It has been checked also that there is no noticeable difference between the two real gas models in terms of computational resources (CPU time and required memory size). Thus, they can be used confidently for supersonic ejector applications with R134a. For sake of clarity, only the REFPROP 7.0 database equation will be considered in the following. For other ejector designs and other operating conditions, this model should nevertheless be used with a certain caution as the temperature range covered by this database (170–455 K) could be easily exceeded during the convergence process, which would cause the solution to fall in regions where the fluid properties are not defined. It may be particularly true during the first statistical iterations where the calculations are highly oscillating if initial conditions are not carefully chosen. Using the REFPROP 7.0 database, Mazzelli and Milazzo (2015) obtained a very low convergence rate of the solution and a weak stability for a supersonic ejector working with R245fa recommending the use of the Peng–Robinson model.

3.2. Influence of the RANS turbulence model

Four two-equation turbulence models described in Section 2.4 have been compared to the experimental data of García del Valle et al. (2014). For all calculations, the real gas model based on the REFPROP 7.0 database has been used.

Table 4 shows the relative deviation $\epsilon = |\omega_{pred} - \omega_{exp}| / \omega_{exp}$ between the predicted ω_{pred} and the experimental $\omega_{exp} = 0.398$ values of the entrainment ratio for OP2. The standard $k-\epsilon$ model with standard wall functions performs the best with a deviation of 4.27%. This model is indeed known to predict well wall-bounded flows with relatively smooth mean pressure gradients, which is the case in the region confined between the primary nozzle exit and the appearance of the shock wave in the mixing chamber. The advanced versions of the $k-\epsilon$ model (realizable $k-\epsilon$ and $k-\epsilon$ RNG) do not improve the predictions of the entrainment ratio. By comparing the low- and high-Reynolds number $k-\omega$ SST models, it is shown that modeling the flow

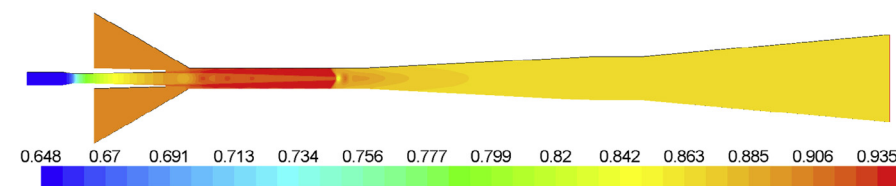


Fig. 5 – Map of the compressibility ratio Z obtained by the standard $k-\epsilon$ model with the REFPROP 7.0 database for OP2.

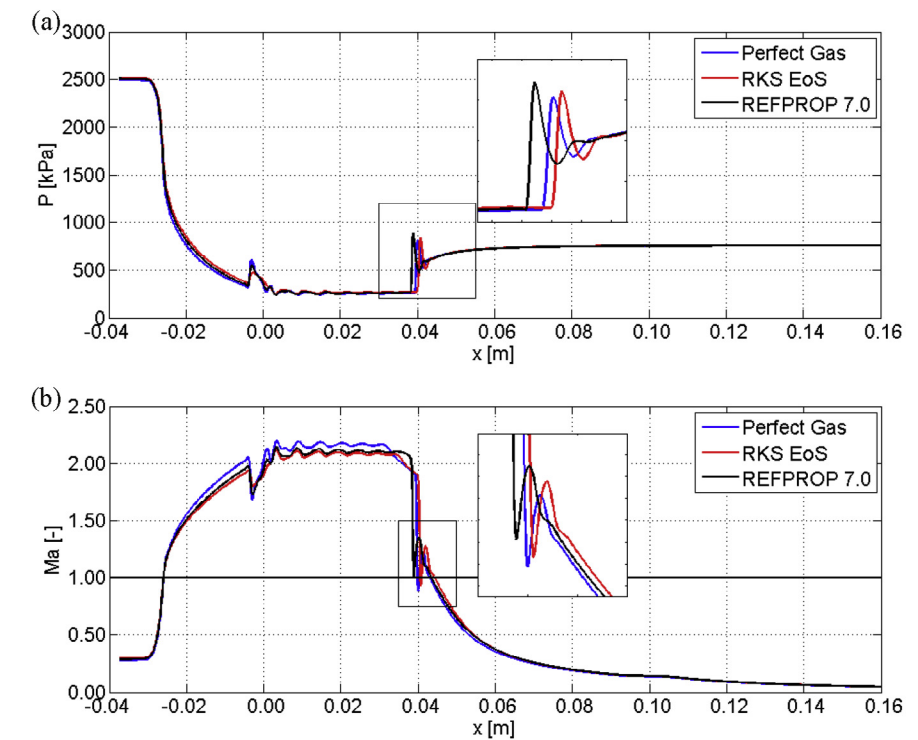


图4 – OP2气体模型比较: (a) 沿引射器中心线的静压分布 P (kPa) 和 (b) 马赫数 Ma 分布。结果采用标准 $k-\epsilon$ 模型获得。

还验证两种真实气体模型在计算资源 (CPU时间和所需内存大小) 方面没有显著差异。因此, 它们可放心用于R134a超音速喷射器应用。为清晰起见, 后续仅考虑REFPROP 7.0数据库方程。对于其他喷射器设计和其他操作条件, 该模型仍应谨慎使用, 因为该数据库覆盖的温度范围 (170–455 K) 在收敛过程中可能轻易被超出, 导致解落入流体属性未定义的区域。尤其在第一统计迭代中, 若初始条件选择不当, 计算高度振荡时更易发生。使用REFPROP 7.0数据库, Mazzelli and Milazzo (2015) 对以R245fa为工质的超音速喷射器获得了非常低的收敛率和较差的稳定性, 建议使用Peng–Robinson模型。

3.2. RANS湍流模型的影响

第2.4节描述的四个双方程湍流模型与García del Valle等人 (2014) 的实验数据进行了比较。所有计算均使用了基于REFPROP 7.0数据库的真实气体模型。

表4显示了预测的 $\epsilon = |\omega_{pred} - \omega_{exp}| / \omega_{exp}$ 与实验的 ω_{pred} 之间的相对偏差 $\omega_{exp} = 0.398$ 。OP2的引射比值。采用标准壁面函数的 $k-\epsilon$ 模型表现最佳, 偏差为4.27%。该模型确实已知能很好预测具有相对平滑平均压力梯度的壁面限制流动, 这正是主喷嘴出口与混合室内激波出现之间区域的典型情况。 $k-\epsilon$ 模型 (可实现型 $k-\epsilon$ 和 $k-\epsilon$ RNG) 的先进版本并未改善引射比的预测。通过比较低雷诺数和高雷诺数 $k-\omega$ SST模型, 表明将流动建模

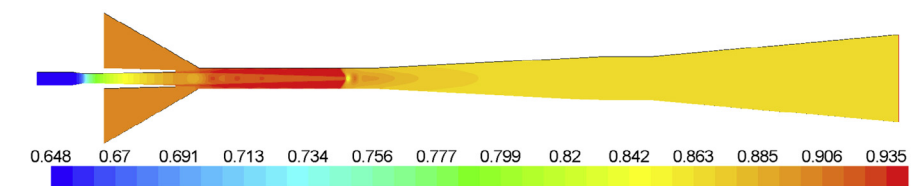


图5 – 协同作用图 的可压缩性比 Z 分布图

Table 4 – Relative deviations ε of the numerical values from the experimental data of García del Valle et al. (2014) in terms of the entrainment ratio for OP2 ($\omega_{exp} = 0.398$). Comparisons between the four two-equation models used in their high- (HRN) or low-Reynolds number (LRN) formulation.

Turbulence model	Relative deviation ε (%)
standard $k-\varepsilon$ (HRN)	4.27
realizable $k-\varepsilon$ (HRN)	8.79
$k-\varepsilon$ RNG (HRN)	6.03
$k-\omega$ SST (HRN)	6.78
$k-\omega$ SST (LRN)	5.70

up to the walls slightly improves the predictions of the entrainment ratio. The present results confirm those obtained by Hemidi et al. (2009a), who showed that the standard $k-\varepsilon$ model in its high-Reynolds number formulation performs better than the high-Reynolds number $k-\omega$ SST model.

Figure 6 compares the numerically predicted operation curves obtained by three turbulence models to the experimental data reported by García del Valle et al. (2014). There is a good overall agreement between the standard $k-\varepsilon$, the $k-\omega$ SST in its low-Reynolds number approach and the experimental results. The difference is around 4% along the on-design conditions ($T_{sat} < 33^\circ$). According to these models, the critical operation point is $T_{sat} = 33^\circ$, which marginally differs from the experimental value $T_{sat} = 32.5^\circ$. Off-design conditions operate up to approximately $T_{sat} = 36^\circ$, which corresponds to the breakdown pressure and to the beginning of the ejector malfunction region. As a conclusion, the low-Reynolds number approach significantly improves the predictions of the entrainment ratio for a given $k-\omega$ SST model, reproducing well the ejector global

performance. In other ejector designs and for other operating conditions, Hemidi et al. (2009a) reported a deviation of 10% for a supersonic air ejector and Sriveerakul et al. (2007) obtained deviations up to 12.9%.

Both the standard $k-\varepsilon$ and low-Reynolds number $k-\omega$ SST models predict quite well the same entrainment ratio over the whole operation curve. Nevertheless, as shown by Hemidi et al. (2009b), the local flow features and especially the shock wave structure may differ significantly. Figure 7 displays the static pressure profile and Mach number distribution along the centerline of the ejector for OP2. Both graphs clearly show that a normal shock wave appears in the mixing chamber, around $x = 0.035$ m for the standard $k-\varepsilon$ model. The other $k-\varepsilon$ models predict the same normal shock wave but for a position slightly downstream, around $x = 0.03$ m for the $k-\varepsilon$ RNG model and $x = 0.0325$ m for the realizable $k-\varepsilon$ model. On the other hand, the $k-\omega$ SST predicts a series of oblique shock waves further downstream in this region, around $x = 0.024$ m. Information about the nature of the shock wave is relatively limited in the literature. Though having considered different ejector geometries and refrigerants, the experimental flow visualizations of Bouhangel and Desevaux (2011), the density field observations of Zhu and Jiang (2014) or the numerical simulations of Bartosiewicz et al. (2006) and Scott et al. (2011) all report a train of oblique shock waves. The low-Reynolds number $k-\omega$ SST model predicts a shock train with a relatively lower intensity as the amplitude of the pressure and Mach number oscillations are smaller than those predicted by the high-Reynolds number approach.

Figure 8 displays the iso-contours of the Mach number in the mixing region at the critical conditions for three turbulence models. It confirms that the standard $k-\varepsilon$ model predicts a single normal shock wave occurring toward the end of the mixing

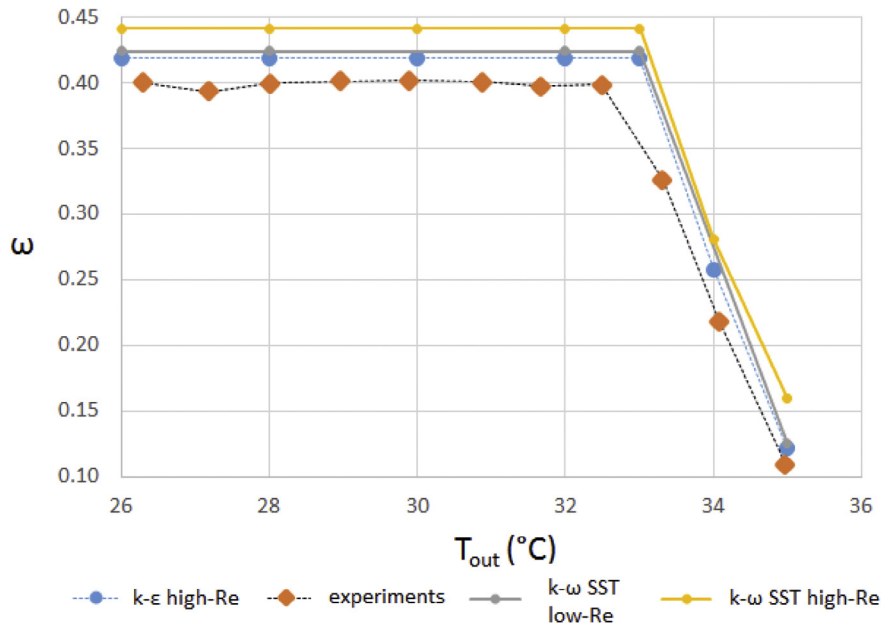


Fig. 6 – Ejector operation curve for a saturation temperature at the primary inlet equal to $T_{sat} = 84.39^\circ\text{C}$. Comparisons between three turbulence models and the experimental data of García del Valle et al. (2014). Results obtained using the REFPROP 7.0 database equation.

表4 – 数值结果相对于García del Valle等人（2014）实验数据的相对偏差 ε ，以OP2的引射比表示（ $\omega_{exp} = 0.398$ ）。比较了四种双方程模型在其高雷诺数（HRN）或低雷诺数（LRN）公式中的结果。

湍流模型	相对偏差 ε (%)
标准 $k-\varepsilon$ (HRN)	4.27
可实现 $k-\varepsilon$ (HRN)	8.79
$k-\varepsilon$ RNG (高雷诺数)	6.03
$k-\omega$ SST (高雷诺数)	6.78
$k-\omega$ SST (低雷诺数)	5.70

至壁面会略微改善引射比的预测。目前的结果证实了Hemidi等人（2009a）的发现，他们表明标准 $k-\varepsilon$ 模型以其高雷诺数公式的表现优于高雷诺数 $k-\omega$ SST模型。

图6比较了三种湍流模型数值预测的运行曲线与García del Valle等人（2014）报告的实验数据。标准 $k-\varepsilon$ 、 $k-\omega$ SST模型及其低雷诺数方法与实验结果总体吻合良好。在设计工况下（ $T_{sat} < 33^\circ$ ），差异约为4%。根据这些模型，临界运行点为 $T_{sat} = 33^\circ$ ，与实验值 $T_{sat} = 32.5^\circ$ 略有差异。非设计工况运行至大约 $T_{sat} = 36^\circ$ ，对应击穿压力和喷射器故障区域的开始。结论是，低雷诺数方法显著改进了给定 $k-\omega$ SST模型对引射比的预测，很好地再现了引射器的整体

性能。在其他喷射器设计和运行条件下，Hemidi等人（2009a）报告了超音速空气引射器的偏差为10%，而Sriveerakul等人（2007）获得的偏差高达12.9%。

标准 $k-\varepsilon$ 和低雷诺数 $k-\omega$ SST模型在整个运行曲线上预测的引射比相当吻合。然而，如Hemidi等人（2009b）所示，局部流动特征尤其是激波结构可能存在显著差异。图7显示了OP2沿喷射器中心线的静压分布和马赫数分布。两张图都清楚地表明，在混合室中出现了正激波，对于标准 $k-\varepsilon$ 模型，位置在 $x = 0.035$ m附近。其他 $k-\varepsilon$ 模型预测了相同的正激波，但位置略向下游，对于 $k-\varepsilon$ RNG模型在 $x = 0.03$ m附近，对于可实现的 $k-\varepsilon$ 模型在 $x = 0.0325$ m附近。另一方面， $k-\omega$ SST模型预测在此区域更下游处有一系列斜激波，位置在 $x = 0.024$ m附近。关于激波特性的文献相对有限。尽管考虑了不同的引射器几何结构和制冷剂，Bouhangel和Desevaux（2011）的实验流动可视化、Zhu和Jiang（2014）的密度场观测，以及Bartosiewicz等人（2006）和Scott等人（2011）的数值模拟都报告了斜激波串。低雷诺数 $k-\omega$ SST模型预测的激波串强度相对较低，因为压力和马赫数振荡的幅度小于高雷诺数方法预测的结果。

图8显示了临界条件下三种湍流模型混合区域内马赫数的等值线。它证实了标准 $k-\varepsilon$ 模型预测在混合室末端附近出现单一正激波

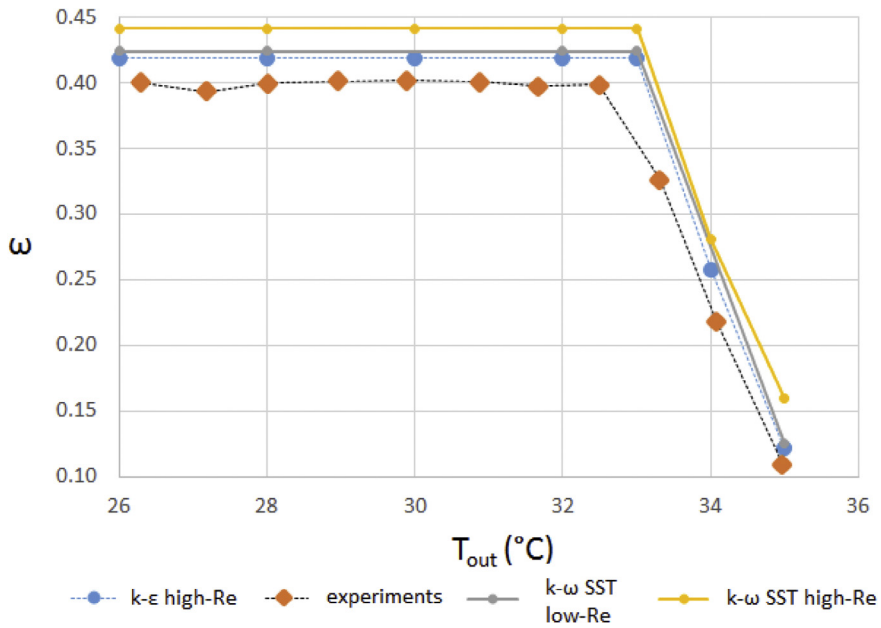


图6 – 主入口饱和温度为 $T_{sat} = 84.39^\circ\text{C}$ 时引射器操作曲线。三种湍流模型与García del Valle等人（2014）实验数据的比较。结果使用REFPROP 7.0数据库方程获得。

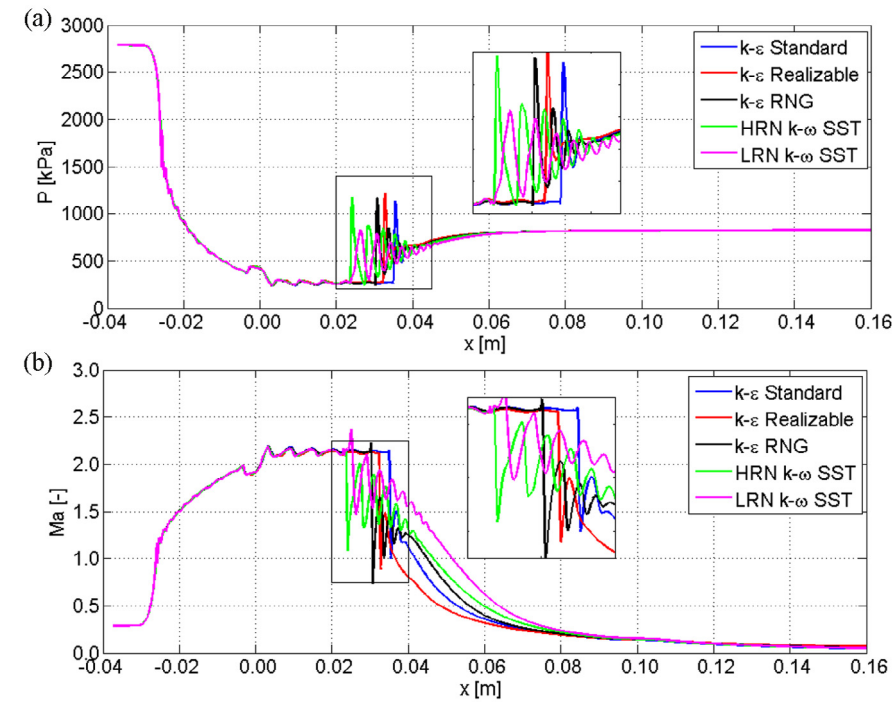


Fig. 7 – Comparisons of the turbulence models for OP2: (a) static pressure profile P (kPa) and (b) Mach number Ma distribution along the centerline of the ejector. Results obtained using the REFPROP 7.0 database equation. HRN (resp. LRN) refers to a high- (resp. low-) Reynolds number approach.

chamber. On the other hand, the low- and high-Reynolds number $k-\omega$ SST models predict the existence of a shock train in the second half of the mixing chamber. Nevertheless, the high-Reynolds number $k-\omega$ SST model slightly anticipates the formation of the shock train. It is interesting to note that the good capture of the shock train structure is inherent to the

turbulence model as both $k-\omega$ SST with high- or low-Reynolds number formulations provide the same results, while very different mesh grid arrangements are used. It is not so surprising as the $k-\omega$ SST uses a $k-\omega$ formulation in the inner parts of the boundary layer, which makes the model directly usable all the way down to the wall through the viscous sublayer. It has been

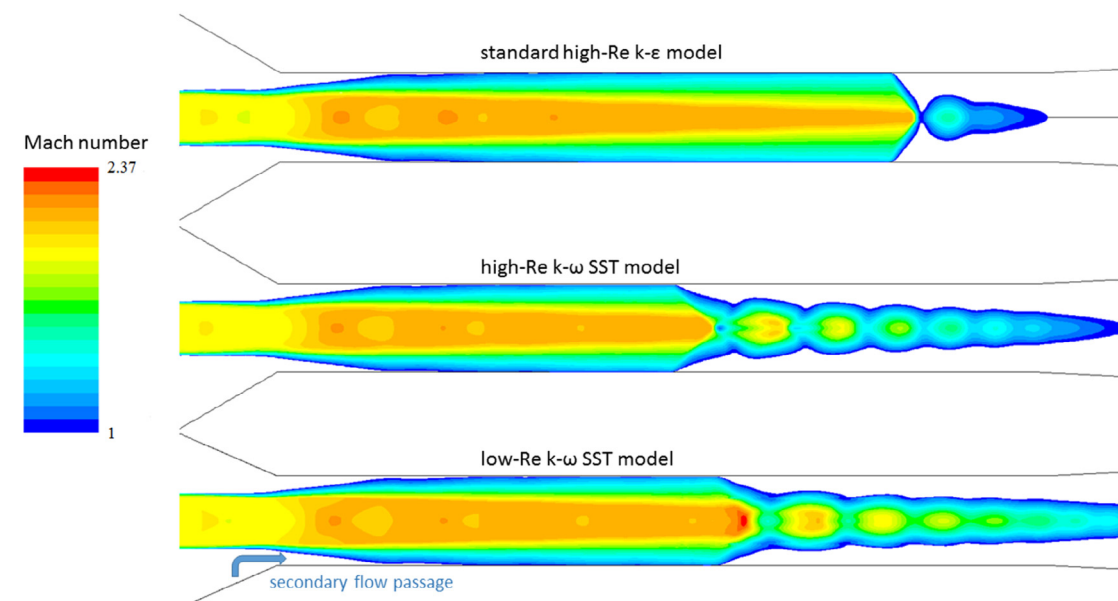


Fig. 8 – Iso-contours of the Mach number in the mixing region for OP2. Comparisons between the standard $k-\epsilon$ and the high- and low-Reynolds number $k-\omega$ SST models.

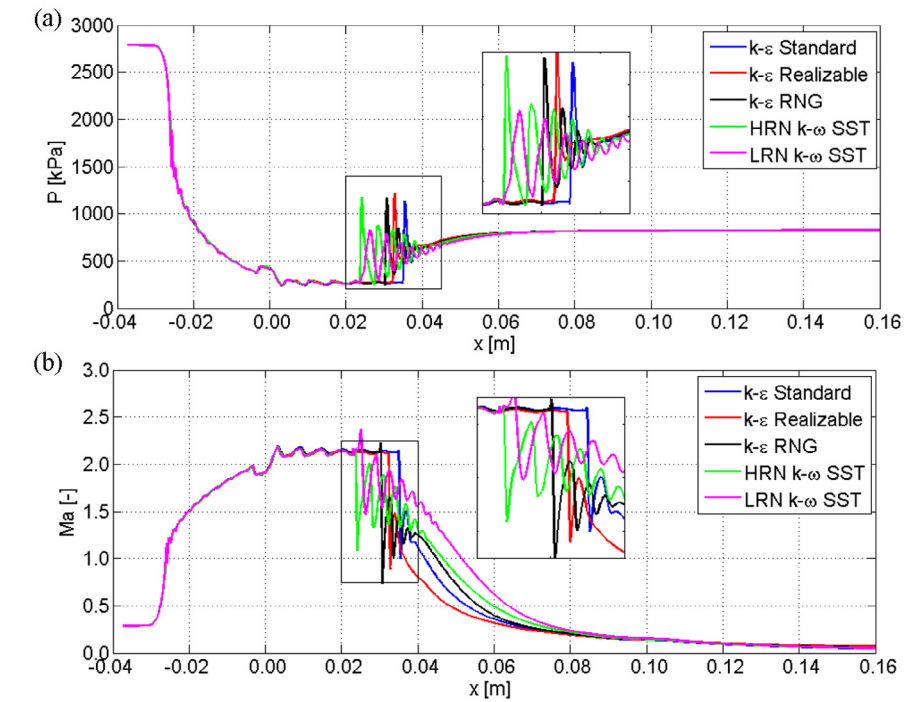


图7 – OP2的湍流模型比较: (a) 静压分布 P (kPa) 和 (b) 沿引射器中心线的马赫数 Ma 分布。结果使用 REFPROP 7.0数据库方程获得。HRN (分别对应LRN) 指高 (分别对应低) 雷诺数方法。

。另一方面, 低雷诺数和高雷诺数 $k-\omega$ SST模型预测在混合室后半部分存在激波串。尽管如此, 高雷诺数 $k-\omega$ SST模型略微提前预测了激波串的形成。有趣的是, 激波串结构的良好捕捉是

湍流模型中, $k-\omega$ SST采用高雷诺数或低雷诺数公式均能得到相同结果, 尽管使用了完全不同的网格布置。这并不令人惊讶, 因为 $k-\omega$ SST在边界层内部使用 $k-\omega$ 公式, 使得模型可直接从主流区域一直应用到壁面并通过黏性底层。

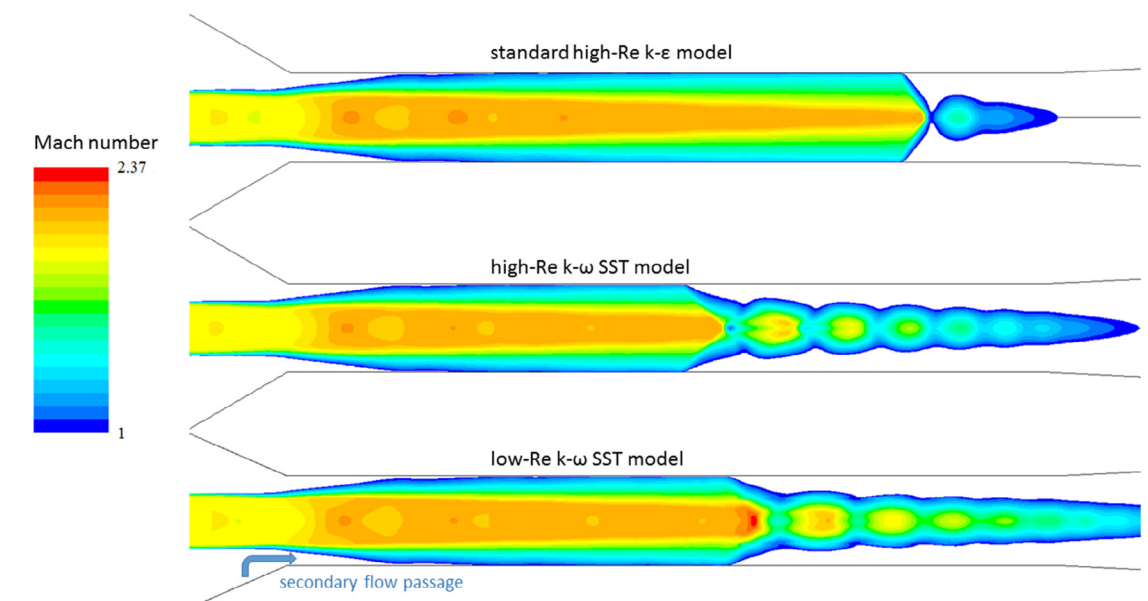


图8 – OP2混合区域中的马赫数等值线。标准 $k-\epsilon$ 与高、低雷诺数 $k-\omega$ SST模型的比较。

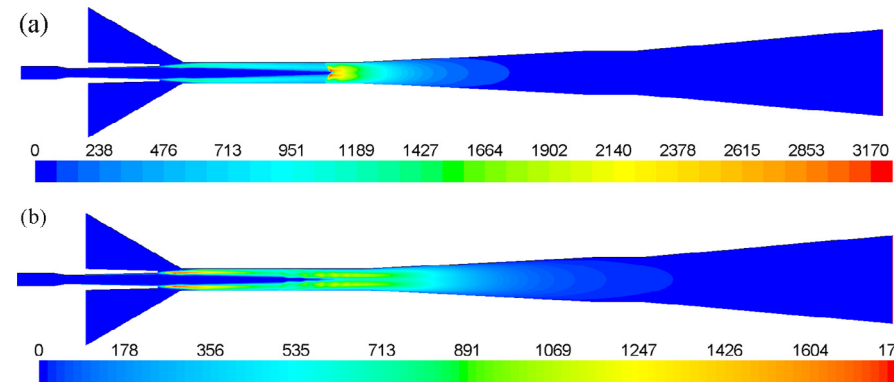


Fig. 9 – Maps of the turbulence kinetic energy k (m^2/s^2) for OP2: (a) standard $k-\epsilon$ and (b) $k-\omega$ SST in their high-Reynolds number formulation. Results obtained using the REFPROP 7.0 database equation.

designed as a low-Reynolds number turbulence model without any extra damping functions even its high-Reynolds number formulation. The SST formulation also switches to a $k-\epsilon$ behavior in the free-stream and thereby avoids the common $k-\epsilon$ problem that the model is too sensitive to the inlet free-stream turbulence properties. The $k-\omega$ SST low-Reynolds number combines then the good prediction of the shock-structure by the high-Reynolds number $k-\omega$ SST model and of the entrainment ratio by the high-Reynolds number $k-\epsilon$ model.

To go a little further and try to explain the different behaviors between these two-equation models, Fig. 9 shows the contours of the turbulence kinetic energy k (m^2/s^2) obtained by the standard $k-\epsilon$ and $k-\omega$ SST models in their high-Reynolds number formulation for OP2. The maps obtained by the other models exhibit quite comparable results and are then not shown here. Turbulence kinetic energy reaches particularly high values in two regions within the ejector: the **shear-layer**, which develops at the trailing edge of the primary nozzle exit and at the exact location of the shock wave in the **constant section area**. Within the **shear-layer**, the maximum values of k are equal to 1475 and 1780 m^2/s^2 for the standard $k-\epsilon$ and $k-\omega$ SST models respectively. This maximum value gets 1690 and 2440 m^2/s^2 for the RNG and realizable models respectively. The high values of the turbulence kinetic energy in the **shear-layer** are directly connected to the high values of the entrainment ratio. At the shock wave location, the standard $k-\epsilon$ and $k-\omega$ SST models predict very different values for the maximum of the

turbulence kinetic energy $k_{\max}$: $k_{\max} = 3170 \text{ m}^2/\text{s}^2$ for the standard $k-\epsilon$ model and $k_{\max} = 10^3 \text{ m}^2/\text{s}^2$ for the $k-\omega$ SST model.

Another interesting quantity when speaking about turbulence modeling is the turbulence viscosity ratio defined as the ratio of the turbulence viscosity ν_t and the kinematic viscosity ν . As shown in Table 3, the viscosity ν varies within the ejector when using the REFPROP 7.0 database. It has been checked that the same contours are obtained for ν whatever the two-equation model used. The turbulence viscosity ratio is then a direct measure of the turbulence viscosity ν_t and so of the energy dissipated by the turbulence model. Figure 10a and b report the iso-contours of the turbulence viscosity ratio for OP2 using the standard $k-\epsilon$ and $k-\omega$ SST high-Reynolds number models. Both models exhibit the same patterns. The turbulence models act mainly in the mixing layer at the trailing edge of the primary nozzle exit and then at the shock wave location. The maximum turbulence viscosity ratio is observed in all cases downstream of the shock waves at the end of the mixing chamber or at the inlet of the diffuser. It is noteworthy that the standard $k-\epsilon$ model is much more dissipative and especially at the shock wave location.

It is noticeable that the maps of the turbulence kinetic energy and the turbulence viscosity ratio are the same between the two $k-\omega$ SST models with slightly different maximum values. For the low-Reynolds approach, $k_{\max} = 1793 \text{ m}^2/\text{s}^2$ and $(\nu_t/\nu)_{\max} = 16,614$. The structure and the position of the shock wave seem to be better captured by the $k-\omega$ SST model

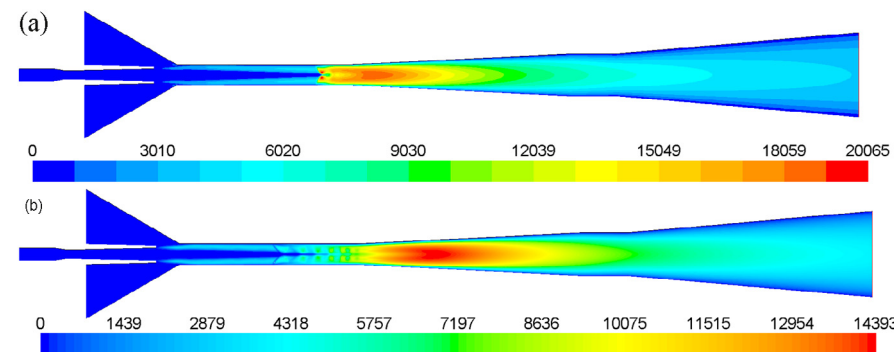


Fig. 10 – Maps of the turbulence viscosity ratio ν_t/ν for OP2: (a) standard $k-\epsilon$ and (b) $k-\omega$ SST in their high-Reynolds number formulation. Results obtained using the REFPROP 7.0 database equation.

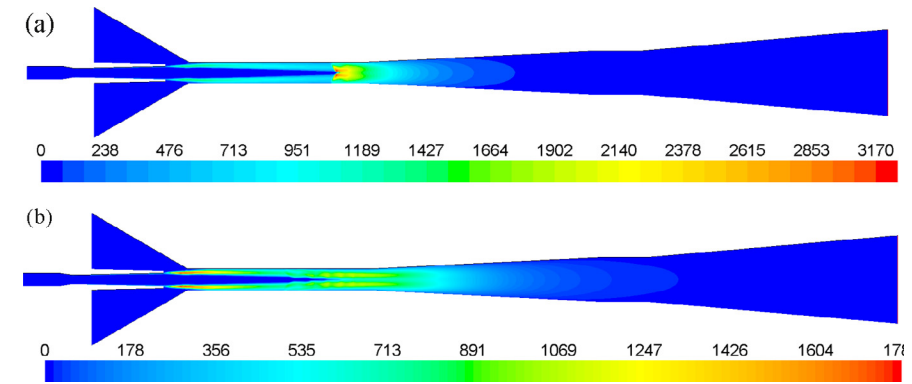


图9 – OP2的湍动能 k (m^2/s^2)分布图: (a)标准 $k-\epsilon$ 和(b) $k-\omega$ SST采用高雷诺数公式。结果使用REFPROP 7.0数据库方程获得。

它被设计为一种无需额外阻尼函数的低雷诺数湍流模型，即使其高雷诺数公式也是如此。SST公式在自由流中也切换为 $k-\epsilon$ 行为，从而避免了常见的 $k-\epsilon$ 问题，即模型对入口自由流湍流特性过于敏感。 $k-\omega$ SST低雷诺数模型结合了高雷诺数 $k-\omega$ SST模型对激波结构的良好预测以及高雷诺数 $k-\epsilon$ 模型对引射比的预测。

为了进一步探讨并解释这两种双方程模型之间的不同表现，图9展示了标准 k^2 和 k^2 SST模型在高雷诺数公式下针对OP2计算得到的湍动能 k (m^2/s^2)的等值线图。其他模型得到的图结果相当，因此在此不展示。湍流动能在引射器内的两个区域达到特别高的值：剪切层（在主喷嘴出口尾缘发展）以及等截面段激波的精确位置。在剪切层内，标准 $k-\epsilon$ 和 $k-\omega$ SST模型的 k 最大值分别为1475和1780 m^2/s^2 ，而RNG和可实现模型的最大值分别为1690和2440 m^2/s^2 。剪切层中高湍流动能值直接与高引射比值相关。在激波位置，标准 $k-\epsilon$ 和 $k-\omega$ SST模型预测的

湍流动能 $k_{\max}$: $k_{\max} = 3170 \text{ m}^2/\text{s}^2$ 对于标准 $k-\epsilon$ 模型和 $k_{\max} = 10^3 \text{ m}^2/\text{s}^2$ 对于 $k-\omega$ SST模型。

谈论湍流建模时另一个有趣的量是湍流黏度比，定义为湍流黏度 ν_t 与运动黏度 ν 之比。如图表3所示，使用REFPROP 7.0数据库时，黏度 ν 在引射器内变化。已证实，无论使用哪种双方程模型，都得到相同的等值线 ν 。湍流黏度比直接衡量湍流黏度 ν_t ，从而衡量湍流模型耗散的能量。图10a和b展示了OP2使用标准 $k-\epsilon$ 和 $k-\omega$ SST高雷诺数模型的湍流黏度比等值线。两种模型呈现相同的模式。湍流模型主要作用于主喷嘴出口后缘的混合层以及激波位置。在所有情况下，最大湍流黏度比出现在混合室末端或扩压器入口处的激波下游。值得注意的是，标准 $k-\epsilon$ 模型耗散性更强，尤其在激波位置。

值得注意的是，两种 $k-\omega$ SST模型的湍动能和湍流黏度比云图相同，仅最大值略有差异。对于低雷诺数方法， $k_{\max} = 1793 \text{ m}^2/\text{s}^2$ 和 $(\nu_t/\nu)_{\max} = 16,614$ 。激波的结构和位置似乎被 $k-\omega$ SST模型更好地捕捉到。

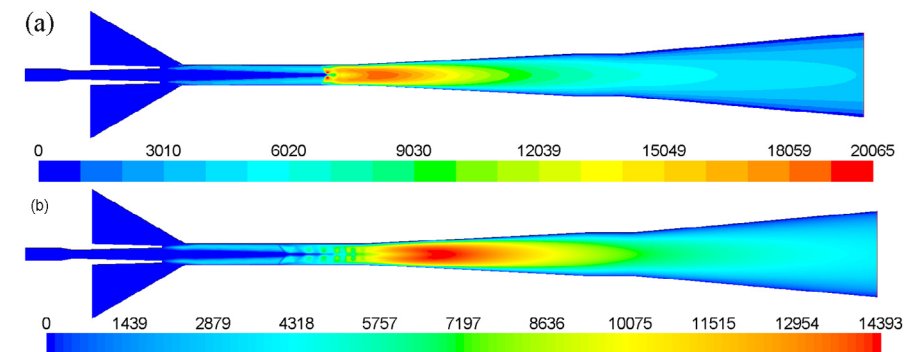


图10 – OP2的湍流黏度比 ν_t/ν 分布图: (a)标准 $k-\epsilon$ 和(b) $k-\omega$ SST采用高雷诺数公式。结果使用REFPROP 7.0数据库方程获得。

either in its high- or low-Reynolds number formulation. The low-Reynolds approach improves the predictions of the flow field within the boundary layers. The secondary fluid passage being confined close to the wall of the mixing chamber, the value of the predicted entrainment ratio is also improved. On the other hand, the advanced versions of the $k-\epsilon$ model do not improve the predictions of the standard model both in terms of the global performance and the more local flow features within the ejector.

4. Conclusion

The flow structure and performance characteristics of a supersonic ejector working with R134a have been studied by 2D axisymmetric, steady-state numerical simulations and compared to the experimental data of García del Valle et al. (2014). A numerical benchmark of some thermodynamic and two-equation turbulence models available within Fluent v.15 have been performed to build the CFD model offering the best compromise between accuracy and computational efforts. The following main conclusions may be drawn:

- The two real gas equations, the Redlich–Kwong–Soave equation of state and the REFPROP 7.0 database equation, provide similar and good results, while the perfect gas law fails to predict the correct entrainment ratio with an overprediction of about 20% whatever the operating conditions.
- The standard $k-\epsilon$ in its high-Reynolds number formulation provides the best agreement in terms of entrainment ratio with a deviation of less than 1% from the experimental value for OP1. This deviation slightly increases when the outlet pressure increases. On the other hand, it predicts a normal shock wave in the mixing chamber instead of an oblique shock train, which is better captured by the $k-\omega$ SST model either in its high- or low-Reynolds number formulation. The maximal deviation for these three models is 4% in terms of entrainment ratio for the three operating conditions.
- Though requiring a calculation time three times longer, the $k-\omega$ SST model in its low-Reynolds number formulation offers the best overall agreement, being capable of capturing the detailed structure of the shock train. As 3D calculations are not required, it appears as a very promising tool even from an engineering point of view to design future ejectors as converged states are achieved within 3 hours with standard computational resources.

The CFD model based on the $k-\omega$ SST model in its low-Reynolds number formulation and the REFPROP 7.0 database equation can be now used confidently to investigate, in the second part of this work, the local flow features and the exergy destruction within the ejector and discuss the different assumptions made by 1D thermodynamic models.

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无论是在高雷诺数还是低雷诺数公式中。低雷诺数方法改善了边界层内流场的预测。由于二次流体通道靠近混合室壁面，预测的引射比值也有所提高。另一方面， $k-\epsilon$ 模型的高级版本并未在整体性能及引射器内部更局部的流动特征方面改善标准模型的预测。

4. 结论

通过二维轴对称稳态数值模拟研究了R134a超音速引射器的流动结构和性能特性，并与García del Valle等人(2014)的实验数据进行了比较。对Fluent v.15中可用的几种热力学和双方程湍流模型进行了数值基准测试，以构建在精度和计算工作量之间取得最佳平衡的CFD模型。可得出以下主要结论：

- 两种真实气体方程——Redlich–Kwong–Soave状态方程和REFPROP 7.0数据库方程——提供了相似且良好的结果，而理想气体定律无法预测正确的引射比，无论操作条件如何，其高估幅度约为20%。
- 标准 $k-\epsilon$ 在高雷诺数公式下，在引射比方面提供了最佳吻合度，对于OP1与实验值的偏差小于1%。当出口压力增加时，该偏差略有增加。另一方面，它预测混合室中为正激波而非斜激波串，而 $k-\omega$ SST模型无论是在高雷诺数还是低雷诺数公式下都能更好地捕获斜激波串。对于这三个模型，在三种操作条件下，引射比的最大偏差为4%。
- 虽然需要三倍的计算时间，但采用低雷诺数公式的 $k-\omega$ SST模型在整体上吻合最佳，能够捕捉激波串的详细结构。由于无需三维计算，即使从工程角度来看，它也是一种非常有前景的工具，可在3小时内借助标准计算资源达到收敛状态，用于设计未来的喷射器。

基于采用低雷诺数公式的 $k-\omega$ SST模型和REFPROP 7.0数据库方程的CFD模型，现可自信地用于研究本文第二部分中喷射器内的局部流动特征和火用损毁，并讨论一维热力学模型所做的不同假设。

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